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Cladribine for people with multiple sclerosis (Review)

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TABLE OF CONTENTS

ABSTRACT	1
PLAIN LANGUAGE SUMMARY	3
SUMMARY OF FINDINGS	5
BACKGROUND	12
OBJECTIVES	12
METHODS	12
RESULTS	15
Figure 1.	16
Figure 2.	20
Figure 3.	21
Figure 4.	22
Figure 5.	23
Figure 6.	24
Figure 7.	24
Figure 8.	25
DISCUSSION	26
AUTHORS' CONCLUSIONS	28
SUPPLEMENTARY MATERIALS	28
ADDITIONAL INFORMATION	28
REFERENCES	30
ADDITIONAL TABLES	35

[Intervention Review]

Cladribine for people with multiple sclerosis

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ABSTRACT

Rationale

Multiple sclerosis (MS) is a chronic, degenerative, autoimmune-mediated disease of the central nervous system (CNS). Although its pathogenesis is not fully understood, inflammation involving both T and B cells is considered an essential factor. Cladribine induces targeted lymphocyte depletion in the peripheral and central nervous system, potentially lowering inflammation and slowing disease progression.

Objectives

To assess the short- and long-term benefits and harms of cladribine compared to no intervention, placebo, or any other disease-modifying drugs (DMDs) in people with any form of multiple sclerosis (MS).

Search methods

We searched the Cochrane MS and Rare Diseases of the CNS Trials Register (part of CENTRAL), MEDLINE, Embase, CINAHL, LILACS, major trials registries, and conference abstracts up to 3 February 2025.

Eligibility criteria

We included randomised controlled trials (RCTs), their open-label extension trials (OLEs), the first phase of cross-over trials, and non-randomised studies of interventions (NRSIs) that investigated cladribine, alone or in combination, at any dose or duration versus no intervention, placebo, or any other DMDs. Participants had a confirmed diagnosis of MS or clinically isolated syndrome according to accepted criteria. We excluded retrospective studies and studies without a comparison group.

Outcomes

Critical outcomes

Number of participants at the end of follow-up: with at least one serious adverse event (SAE), free from disability worsening, with no evidence of disease activity (NEDA), who withdrew from treatment, and with cognitive impairment; and quality of life.

Important outcomes

Number of participants at the end of follow-up with: at least one adverse event, at least one new relapse, and new gadolinium-enhancing positive T1-weighted (Gd+ T1) or new/enlarging T2-weighted magnetic resonance imaging (MRI) lesions.

Risk of bias

We used the Cochrane risk of bias tools RoB 2 (for RCTs and OLEs) and ROBINS-I (for NRSIs).

Synthesis methods

We conducted random-effects meta-analyses to calculate risk ratios (RRs) with 95% confidence intervals (CIs), assessing RCTs, OLEs, and NRSIs separately. We assessed the certainty of evidence using GRADE.

Included studies

We included 15 studies: nine RCTs (follow-up 52–96 weeks; 2571 participants), two medium-term follow-up OLEs (24–96 weeks; 915 participants), two long-term follow-up OLEs (9.5–11.4 years), one NRSI with one year of follow-up (37 participants in cladribine group vs 599 in interferon beta [IFN- β], fingolimod, and natalizumab groups), and one NRSI with three years of follow-up (633 participants in cladribine group vs 2842 in fingolimod, dimethyl fumarate, and teriflunomide groups). Three reports assessed MRI outcomes. No studies reported quality of life or cognitive impairment.

Synthesis of results

Randomised controlled trials

Cladribine compared to placebo may have little to no effect on the risk of SAEs (RR 1.08, 95% CI 0.80 to 1.45; 8 studies, 2495 participants) and treatment withdrawal (RR 1.43, 95% CI 0.77 to 2.66; $I^2 = 76\%$; 8 studies, 2503 participants), but the evidence for both outcomes is very uncertain. Cladribine may have little to no effect on the number of people remaining free from disability worsening up to 24 months (RR 1.05, 95% CI 0.96 to 1.16; 3 studies, 1549 participants; low certainty).

Cladribine compared to placebo may increase the risk of any adverse events, but the evidence is very uncertain (RR 1.14, 95% CI 1.02 to 1.28; 8 studies, 2405 participants). Cladribine likely reduces the risk of new relapses (RR 0.53, 95% CI 0.45 to 0.62; 2 studies, 1498 participants; moderate certainty), and it may reduce the risk of new Gd+ T1 MRI lesions, although the results are heterogeneous and the evidence is very uncertain (RR 0.30, 95% CI 0.19 to 0.47; $I^2 = 79\%$; 4 studies, 2153 participants).

Open-label extension trials

With follow-up beyond two years, cladribine compared to placebo may have little to no effect on the risk of SAEs (RR 0.89, 95% CI 0.60 to 1.33; 2 studies, 915 participants), freedom from disability worsening (RR 1.02, 95% CI 0.93 to 1.11; 1 study, 806 participants), and treatment withdrawal (RR 0.93, 95% CI 0.81 to 1.08; 2 studies, 915 participants), but the evidence is very uncertain for all three outcomes. Cladribine may increase the proportion of people with NEDA at four years (RR 1.63, 95% CI 1.23 to 2.16; 2 studies, 662 participants; low certainty).

Cladribine compared to placebo may have little to no effect on the risk of any adverse events (RR 1.04, 95% CI 0.96 to 1.12; 2 studies, 916 participants) and may reduce relapses at long-term follow-up (RR 0.68, 95% CI 0.59 to 0.79; 2 studies, 662 participants), but the evidence is very uncertain for both outcomes. In one study, cladribine reduced new Gd+ T1 MRI lesions, but the evidence is very uncertain (RR 0.68, 95% CI 0.52 to 0.88; 806 participants).

Non-randomised studies of interventions

Observational comparisons (1 study) showed reductions in relapse over three years with cladribine versus:

- dimethyl fumarate (RR 0.57, 95% CI 0.38 to 0.88; low certainty);
- teriflunomide (RR 0.30, 95% CI 0.20 to 0.47; moderate certainty); and
- fingolimod (RR 0.51, 95% CI 0.36 to 0.74; moderate certainty).

However, the evidence is very uncertain for the effect of cladribine on relapse over 12 months versus:

- fingolimod (RR 1.05, 95% CI 0.52 to 2.10; very low certainty);
- natalizumab (RR 1.25, 95% CI 0.58 to 2.71; very low certainty); and
- interferon beta (RR 0.65, 95% CI 0.26 to 1.62; very low certainty).

Authors' conclusions

Cladribine is an oral treatment for adults with MS requiring few monitoring visits and only short courses over two years, thereby reducing treatment burden, with a safety profile that appears acceptable based on very low-certainty evidence. Evidence from RCTs suggests cladribine likely reduces new relapses, may reduce new Gd+ T1 MRI lesions (although the evidence is very uncertain), but may have little to no effect on slowing disability progression. Results from OLEs showed cladribine may increase the proportion of people with NEDA at four years. Benefits in subgroups and for some critical outcomes remain uncertain due to the paucity of studies. Further high-quality controlled trials and patient-centred pragmatic registries are needed to strengthen the evidence base.

Funding

This Cochrane review had no dedicated funding.

Registration

Protocol available via <https://doi.org/10.1002/14651858.CD013524>.

PLAIN LANGUAGE SUMMARY

Is cladribine effective and safe for treating people with multiple sclerosis (MS)?

Key messages

- Compared with placebo, cladribine likely reduces the risk of relapse, and it may reduce the number of new lesions seen on brain scans in people with multiple sclerosis after almost two years of treatment, but the evidence is very uncertain. Cladribine may result in little to no difference in slowing the progression of disability after almost two years of treatment. Compared with placebo, cladribine may not change the risk of serious side effects, but the evidence is very uncertain.
- Longer and better-quality studies are needed to assess the benefits and harms of cladribine acting on the immune system in multiple sclerosis. Future research should compare cladribine with other medicines used in this disease, focusing on effects that are most important to patients, such as quality of life and the ability to complete everyday tasks. Future studies should also include vulnerable populations.

What is multiple sclerosis?

Multiple sclerosis (MS) is an uncommon disease caused by inflammation of the brain and spinal cord, leading to damage that impairs important everyday activities over time, such as walking and self-care. People with MS experience symptoms such as visual disturbances, numbness/tingling, weakness, bladder and bowel dysfunction, problems with mental abilities, fatigue, and dizziness. These symptoms may worsen and lead to dependence upon others. The reappearance of symptoms is called 'relapse'. The most common form of MS is called 'relapsing-remitting MS', where symptoms come and go. Over time, relapses become more frequent, with more troublesome symptoms and shorter periods of well-being in between. MS is a particularly burdensome condition; it typically affects young people, mainly women, in the most active stage of their lives.

How is multiple sclerosis treated?

MS is currently treated with 'disease-modifying' medicines (e.g. interferon beta, glatiramer acetate, natalizumab, fingolimod, alemtuzumab, teriflunomide, and dimethyl fumarate). These medicines can slow but not stop disease progression.

What did we want to find out?

The exact causes of MS are not fully understood, but we know that inflammation, driven by certain immune cells (T and B cells), plays an important role. Cladribine works by reducing some of these immune cells in the blood and brain, which may help lower inflammation and slow down the course of the disease. We wanted to know if cladribine can slow or stop the worsening of disability in people with MS, with no increase in unwanted events.

What did we do?

We searched for studies comparing cladribine with no treatment, placebo (dummy treatment), or any 'disease-modifying' medicines in people with any type of MS. We compared and summarised the results of the studies and rated our confidence in the evidence, based on factors such as study methods, number of participants, and precision of the results.

What did we find?

We included 15 studies with the following designs.

- Nine randomised controlled trials (RCTs, where participants are assigned randomly to two or more treatment groups) that followed 2571 participants for one to two years.
- Four open-label extension trials (OLEs, where participants from RCTs are offered continued access to the study treatment) that followed participants for another two to 10 years.
- Two observational studies (using 'real-life' data) that compared cladribine with other 'disease-modifying' medicines in 4111 people over one to three years.

Important outcomes for patients, such as quality of life and thinking or memory problems, were not reported in any study.

Results from RCTs suggest cladribine probably reduces the risk of new relapses compared to placebo. It may also lower the number of new lesions seen on magnetic resonance imaging (MRI) scans of the brain, although the evidence is very uncertain. However, cladribine may

have little or no effect on whether disability gets worse. There may be little or no difference between cladribine and placebo in the number of people with serious unwanted events and who stop using treatment, and cladribine may increase the risk of other unwanted events, but our confidence in these results is very low.

Results from OLEs suggest cladribine may increase the proportion of people who show no signs of the disease worsening at four years compared to placebo. It may reduce relapses and the number of new MRI lesions, but the evidence is very uncertain. It may have little to no effect on whether disability gets worse, whether people experience unwanted events, or whether they stop using treatment, but we have very little confidence in these results.

From observational studies comparing cladribine with other medicines, results were mixed: cladribine may be better than some medicines (dimethyl fumarate, teriflunomide, and fingolimod at three years) for reducing relapses, but its effects on this outcome compared with fingolimod, natalizumab, and interferon beta at one year remain unclear.

What are the limitations of the evidence?

Our confidence in the effects of cladribine is limited by poor-quality research and loss of focus on affected people's needs. Most studies were performed by the cladribine manufacturing company. There is concern that their interests may have influenced reporting of study results.

How up-to-date is this evidence?

The evidence is current to February 2025.

SUMMARY OF FINDINGS

Summary of findings 1. Summary of findings table - Cladribine compared to placebo for people with multiple sclerosis: results from randomised controlled trials

Cladribine compared to placebo for people with multiple sclerosis: results from randomised controlled trials

Patient or population: people with multiple sclerosis: results from randomised controlled trials

Setting: outpatient

Intervention: cladribine

Comparison: placebo

Outcomes	Anticipated absolute effects* (95% CI)		Relative effect (95% CI)	Nº of participants (studies)	Certainty of the evidence (GRADE)	Comments
	Risk with placebo	Risk with cladribine				
Number of participants with at least 1 serious adverse event (SAE) at the end of follow-up (range 12–22 months)	67 per 1000	73 per 1000 (98 to 54)	RR 1.08 (0.80 to 1.45)	2495 (8 RCTs)	⊕⊕⊕⊕ Very low ^{a,b}	Cladribine may have little to no effect on the risk of experiencing ≥ 1 SAE by the end of follow-up, but the evidence is very uncertain.
Number of participants free from disability worsening measured with the expanded disability status scale (EDSS) at the end of follow-up (range 12–22 months) ^c	796 per 1000	835 per 1000 (923 to 764)	RR 1.05 (0.96 to 1.16)	1549 (3 RCTs)	⊕⊕⊕⊕ Low ^d	Cladribine may have little to no effect on the number of people who are free from disability worsening up to 24 months.
Number of participants with no evidence of disease activity at the end of follow-up - not reported	-	-	-	-	-	No studies reported this outcome.
Number of participants who withdrew from treatment before the end of follow-up (range 12–22 months)	124 per 1000	177 per 1000 (330 to 95)	RR 1.43 (0.77 to 2.66)	2503 (8 RCTs)	⊕⊕⊕⊕ Very low ^{b,e,f}	Cladribine may have little to no effect on the risk of treatment withdrawal before the end of follow-up, but the evidence is very uncertain.
Number of participants with evidence of cognitive impairment at 12 months and at the end of follow-up - not reported	-	-	-	-	-	No studies reported this outcome.

Quality of life - not reported	-	-	-	-	-	No studies reported this outcome.
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***The risk in the intervention group** (and its 95% confidence interval) is based on the assumed risk in the comparison group and the **relative effect** of the intervention (and its 95% CI).

CI: confidence interval; **RR:** risk ratio

GRADE Working Group grades of evidence

High certainty: we are very confident that the true effect lies close to that of the estimate of the effect.

Moderate certainty: we are moderately confident in the effect estimate: the true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different.

Low certainty: our confidence in the effect estimate is limited: the true effect may be substantially different from the estimate of the effect.

Very low certainty: we have very little confidence in the effect estimate: the true effect is likely to be substantially different from the estimate of effect.

See interactive version of this table: https://gdt.grade.pro.org/presentations/#/isof/isof_question_revman_web_433817719118808283.

^a Downgraded 2 levels for very serious risk of bias (6/8 studies had a high overall risk of bias).

^b Downgraded 1 level for serious imprecision due to a wide confidence interval crossing the null.

^c EDSS is an ordinal scale for disability where 0 is normal and 10 indicates death from multiple sclerosis.

^d Downgraded 2 levels for very serious risk of bias (2/3 studies had a high overall risk of bias, with a high risk of bias in 2 domains).

^e Downgraded 2 levels for very serious risk of bias (5/8 studies had high overall risk of bias).

^f Downgraded 1 level for serious inconsistency ($I^2 = 76\%$).

Summary of findings 2. Summary of findings table - Cladribine compared to placebo for people with multiple sclerosis: results from open-label extension trials

Cladribine compared to placebo for people with multiple sclerosis: results from open-label extension trials

Patient or population: people with multiple sclerosis: results from open-label extension trials

Setting: outpatient

Intervention: cladribine

Comparison: placebo

Outcomes	Anticipated absolute effects* (95% CI)		Relative effect (95% CI)	Nº of participants (studies)	Certainty of the evidence (GRADE)	Comments
	Risk with placebo	Risk with cladribine				
Number of participants with at least 1 serious adverse event at	120 per 1000	107 per 1000 (160 to 72)	RR 0.89 (0.60 to 1.33)	915 (2 RCTs)	⊕⊕⊕⊕ Very low ^{a,b,c}	Cladribine may have little to no effect on the risk of experiencing ≥ 1 serious

the end of follow-up (range 96–120 weeks)						adverse event during a follow-up period longer than 2 years, but the evidence is very uncertain.
Number of participants free from disability worsening measured with the expanded disability status scale (EDSS) at the end of follow-up (120 weeks) ^d	753 per 1000	768 per 1000 (835 to 700)	RR 1.02 (0.93 to 1.11)	806 (1 RCT)	⊕⊕⊕⊕ Very low ^{e,f}	Cladribine may have little to no effect on the number of people remaining free from disability worsening by the end of follow-up, but the evidence is very uncertain.
Number of participants with no evidence of disease activity at the end of follow-up (4 years after randomisation)	330 per 1000	538 per 1000 (714 to 406)	RR 1.63 (1.23 to 2.16)	662 (2 RCTs)	⊕⊕⊕⊕ Low	Cladribine may increase the probability of having no evidence of disease activity 4 years after randomisation.
Number of participants who withdrew from treatment before the end of follow-up (range 96–120 weeks)	288 per 1000	268 per 1000 (311 to 233)	RR 0.93 (0.81 to 1.08)	915 (2 RCTs)	⊕⊕⊕⊕ Very low ^{b,f}	Cladribine may have little to no effect on the number of people who withdraw from treatment before the end of follow-up, but the evidence is very uncertain.
Number of participants with no evidence of cognitive impairment at the end of follow-up - not reported	-	-	-	-	-	No studies reported this outcome.
Quality of life - not reported	-	-	-	-	-	No studies reported this outcome.

***The risk in the intervention group** (and its 95% confidence interval) is based on the assumed risk in the comparison group and the **relative effect** of the intervention (and its 95% CI).

CI: confidence interval; **RR:** risk ratio

GRADE Working Group grades of evidence

High certainty: we are very confident that the true effect lies close to that of the estimate of the effect.

Moderate certainty: we are moderately confident in the effect estimate: the true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different.

Low certainty: our confidence in the effect estimate is limited: the true effect may be substantially different from the estimate of the effect.

Very low certainty: we have very little confidence in the effect estimate: the true effect is likely to be substantially different from the estimate of effect.

See interactive version of this table: https://gdt.gradepro.org/presentations/#/isof/isof_question_revman_web_448245639444989613.

^a Downgraded 2 levels for very serious risk of bias (both studies had a high overall risk of bias).

^b Judged to have very serious indirectness, as the 2 population studies involved different interventions. Participants entered in the open-label maintenance period of Leist_OLMP (2014) were all treated with subcutaneous interferon (IFN), whereas participants in Giovannoni (2018) received either cladribine or a placebo.

^c Downgraded 1 level for serious imprecision due to a wide confidence interval crossing the null.

^d EDSS is an ordinal scale for disability where 0 is normal and 10 indicates death from MS.

^e Downgraded 2 levels for very serious risk of bias.

^f Downgraded 1 level for serious imprecision, as no sample size calculation was performed, and enrollment was limited to eligible participants from the original randomized study/studies.

Summary of findings 3. Summary of findings table - Cladribine compared to placebo for people with multiple sclerosis: results from randomised controlled trials (important outcomes)

Patient or population: people with multiple sclerosis: results from randomised controlled trials (RCTs)

Setting: outpatient

Intervention: cladribine

Comparison: placebo

Outcomes	Anticipated absolute effects* (95% CI)		Relative effect (95% CI)	N° of participants (studies)	Certainty of the evidence (GRADE)	Comments
	Risk with placebo	Risk with cladribine				
Number of participants with at least 1 adverse event by the end of follow-up (range 12–22 months)	637 per 1000	726 per 1000 (650 to 815)	RR 1.14 (1.02 to 1.28)	2405 (8 RCTs)	⊕⊕⊕⊕ Very low ^{a,b}	Cladribine may increase the risk of experiencing ≥ 1 adverse event by the end of follow-up, but the evidence is very uncertain.
Number of participants with at least 1 new relapse by the end of follow-up (96 weeks)	400 per 1000	212 per 1000 (180 to 248)	RR 0.53 (0.45 to 0.62)	1498 (2 RCTs)	⊕⊕⊕⊕ Moderate ^c	Cladribine likely reduces the risk of experiencing ≥ 1 new relapse by the end of follow-up.
Number of participants with new Gd+ T1 MRI lesions at the end of follow-up (range 12–22 months)	495 per 1000	153 per 1000 (104 to 223)	RR 0.30 (0.19 to 0.47)	2153 (4 RCTs)	⊕⊕⊕⊕ Very low ^{d,e}	Cladribine may reduce the risk of developing new Gd+ T1 MRI lesions by the end of follow-up, but the evidence is very uncertain.

***The risk in the intervention group** (and its 95% confidence interval) is based on the assumed risk in the comparison group and the relative effect of the intervention (and its 95% CI).

CI: confidence interval; **Gd+ T1 MRI lesions:** gadolinium-enhancing T1-weighted magnetic resonance imaging lesions; **RR:** risk ratio.

GRADE Working Group grades of evidence

High certainty: we are very confident that the true effect lies close to that of the estimate of the effect.

Moderate certainty: we are moderately confident in the effect estimate: the true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different.

Low certainty: our confidence in the effect estimate is limited: the true effect may be substantially different from the estimate of the effect.

Very low certainty: we have very little confidence in the effect estimate: the true effect is likely to be substantially different from the estimate of effect.

^a Downgraded 2 levels for very serious risk of bias (6/8 studies had high overall risk of bias).

^b Downgraded 1 level for serious inconsistency ($I^2 = 53\%$).

^c Downgraded 1 level for serious overall risk of bias (1/2 studies had high overall risk of bias).

^d Downgraded 2 levels for very serious overall risk of bias (3/4 studies had a high overall risk of bias).

^e Downgraded 1 level for serious inconsistency ($I^2 = 73\%$).

Summary of findings 4. Summary of findings table - Cladribine compared to placebo for people with multiple sclerosis: results from open-label extension trials (important outcomes)

Patient or population: people with multiple sclerosis: results from open-label extension trials (OLEs)

Setting: outpatient

Intervention: cladribine

Comparison: placebo

Outcomes		Anticipated absolute effects* (95% CI)		Relative ef- fect (95% CI)	Nº of partici- pants (studies)	Certainty of the evidence (GRADE)	Comments
		Risk with placebo	Risk with cladribine				
Number of participants with at least 1 adverse event at the end of follow-up (96–120 weeks)		768 per 1000	799 per 1000 (737 to 860)	RR 1.04 (0.96 to 1.12)	915 (2 OLEs)	⊕⊕⊕⊕ Very low ^{a,b}	Cladribine may have little to no effect on the risk of experiencing ≥ 1 adverse event by the end of follow-up, but the evidence is very uncertain.
Number of par- ticipants with at least 1 new re- lapse at the end of follow-up	Follow-up: 120 weeks (27.6 months; 2.3 years)	321 per 1000	276 per 1000 (215 to 353)	RR 0.86 (0.67 to 1.10)	806 (1 OLE)	⊕⊕⊕⊕ Very low ^{c,d}	Cladribine may reduce the risk of experienc- ing ≥ 1 new relapse over 120 weeks, but the evidence is very uncertain.
	Follow-up range 9.5–11.4 years	723 per 1000	492 per 1000 (427 to 571)	RR 0.68 (0.59 to 0.79)	662 (2 OLEs)	⊕⊕⊕⊕ Very low ^{c,d}	Cladribine may reduce the risk of experienc- ing ≥ 1 new relapse over 9.5–11.4 years, but the evidence is very uncertain.

Number of participants with new Gd+ T1 MRI lesions at the end of follow-up (120 weeks; 27.6 months; 2.3 years)	316 per 1000	215 per 1000 (164 to 278)	RR 0.68 (0.52 to 0.88)	806 (1 OLE)	⊕⊕⊕⊕ Very low ^{c,d}	Cladribine may reduce the risk of developing new Gd+ T1 lesions, but the evidence is very uncertain.
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***The risk in the intervention group** (and its 95% confidence interval) is based on the assumed risk in the comparison group and the **relative effect** of the intervention (and its 95% CI).

CI: confidence interval; **RR:** risk ratio; **Gd+ T1 MRI lesions:** gadolinium-enhancing T1-weighted magnetic resonance imaging lesions.

GRADE Working Group grades of evidence

High certainty: we are very confident that the true effect lies close to that of the estimate of the effect.

Moderate certainty: we are moderately confident in the effect estimate: the true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different.

Low certainty: our confidence in the effect estimate is limited: the true effect may be substantially different from the estimate of the effect.

Very low certainty: we have very little confidence in the effect estimate: the true effect is likely to be substantially different from the estimate of effect.

^a Downgraded 2 levels for very serious risk of bias (both studies had high overall risk of bias).

^b Judged to have very serious indirectness, as the 2 population studies involved different interventions: participants in Leist 2014 were all treated with subcutaneous interferon (IFN), while those in Giovannoni 2018 received either cladribine or a placebo.

^c Downgraded 2 levels for very serious risk of bias.

^d Downgraded 1 level for serious imprecision, as no sample size calculation was performed, and enrolment was limited to eligible participants from the original trial.

Summary of findings 5. Summary of findings: cladribine versus other disease-modifying drugs in multiple sclerosis — evidence from non-randomised studies of interventions

Patient or population: people with multiple sclerosis: results from non-randomised studies of interventions

Setting: outpatient

Intervention: cladribine

Comparison: other disease-modifying drugs (DMDs)

Outcome	DMD	Anticipated absolute effects (95% CI)		Relative effect (95% CI)	Nº of participants (studies)	Certainty	What happens
		Risk with other DMDs	Risk with cladribine				
Number of participants with at least 1 new relapse by 12	Fingolimod	209 per 1000	220 per 1000 (109 to 440)	RR 1.05 (0.52 to 2.10)	290 (1 NRSI)	⊕⊕⊕⊕ Very low ^{a,b}	Cladribine compared with fingolimod may have little to no effect on the risk of experiencing ≥ 1 new relapse over 12 months after randomisation, but the evidence is very uncertain.



months after randomisation	Natalizumab	184 per 1000	230 per 1000 (107 to 498)	RR 1.25 (0.58 to 2.71)	200 (1 NRSI)	⊕⊕⊕⊕ Very low ^{a, b}	Cladribine compared with natalizumab may have little to no effect on the risk of experiencing ≥ 1 new relapse over 12 months after randomisation, but the evidence is very uncertain.
	Interferon beta (IFNβ)	281 per 1000	183 per 1000 (73 to 456)	RR 0.65 (0.26 to 1.62)	189 (1 NRSI)	⊕⊕⊕⊕ Very low ^{a, b}	Cladribine compared with IFNβ may have little to no effect on the risk of experiencing ≥ 1 new relapse over 12 months after randomisation, but the evidence is very uncertain.
Number of participants with at least 1 new relapse by 3 years after randomisation	Fingolimod	146 per 1000	75 per 1000 (53 to 108)	RR 0.51 (0.36 to 0.74)	1040 (1 NRSI)	⊕⊕⊕⊖ Moderate ^a	Cladribine compared with fingolimod likely reduces the risk of experiencing ≥ 1 new relapse over 3 years after randomisation.
	Dimethyl fumarate	120 per 1000	68 per 1000 (46 to 106)	RR 0.57 (0.38 to 0.88)	900 (1 NRSI)	⊕⊕⊕⊖ Low ^{a, c}	Cladribine compared with dimethyl fumarate may reduce the risk of experiencing ≥ 1 new relapse over 3 years after randomisation.
	Teriflunomide	179 per 1000	54 per 1000 (36 to 84)	RR 0.30 (0.20 to 0.47)	916 (1 NRSI)	⊕⊕⊕⊖ Moderate ^a	Cladribine compared with teriflunomide likely results in a large reduction in the risk of experiencing ≥ 1 new relapse over 3 years after randomisation.

***The risk in the intervention group** (and its 95% confidence interval) is based on the assumed risk in the comparison group and the **relative effect** of the intervention (and its 95% CI).

CI: confidence interval; **RR:** risk ratio.

GRADE Working Group grades of evidence

High certainty: we are very confident that the true effect lies close to that of the estimate of the effect.

Moderate certainty: we are moderately confident in the effect estimate: the true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different.

Low certainty: our confidence in the effect estimate is limited: the true effect may be substantially different from the estimate of the effect.

Very low certainty: we have very little confidence in the effect estimate: the true effect is likely to be substantially different from the estimate of effect.

^a Downgraded 1 level for serious risk of bias.

^b Downgraded 2 levels for very serious imprecision, because the 95% CI crosses the line of no effect and includes values compatible with both important benefit and harm.

^c Downgraded 1 level for serious imprecision: although the effect favours cladribine, the 95% CI is wide and includes small effects close to no important benefit; moreover, the total number of events is low (85 across groups), below typical optimal information size (OIS) thresholds, leaving appreciable uncertainty about the true magnitude of benefit.

BACKGROUND

Description of the condition

Multiple sclerosis (MS) is a chronic, degenerative, autoimmune-mediated disorder of the central nervous system (CNS), characterised by inflammation, demyelination, axonal loss, and gliosis. Its aetiology is not fully understood, but inflammation involving both T and B cells plays a central role, leading to myelin loss and oligodendrocyte activation [1, 2]. Neurodegeneration, closely linked to inflammation, emerges early and drives disability progression [3, 4].

MS is the most common demyelinating disease, particularly in young adults [5, 6], and occurring two to three times more often in women. Global prevalence is about 36 per 100,000 people [7, 8], with wide geographic variation, from around 2 per 100,000 in Eastern Asia and sub-Saharan Africa to more than 100 per 100,000 in North America and Europe [9].

The disease imposes a major societal and healthcare burden. Symptoms and course vary widely, although the literature describes key phenotypes. Relapsing-remitting MS (RRMS), the most frequent type ($\approx 85\%$), features relapses lasting 24 hours or more followed by remissions. More than half of patients develop secondary progressive MS (SPMS) within 15 years [10, 11]. About 15% have a gradual onset and steady progression (primary progressive MS; PPMS). In 2014, experts recommended dropping the established 'progressive relapsing' category (PRMS) and adopting the term clinically isolated syndrome (CIS) [12], representing a first demyelinating event with potential progression to MS. A subgroup of people with RRMS experiences highly active MS (HAMS), marked by rapid disability accumulation, though definitions and management remain inconsistent [13].

Disability accrual in RRMS occurs not only through incomplete recovery after relapses (relapse-associated worsening; RAW) but also through progression independent of relapse activity (PIRA) [14], now considered the main driver of disability across phenotypes. However, neither RAW nor PIRA are validated as clinical trial outcomes [15].

Prognosis varies by age, sex, MS subtype, magnetic resonance imaging (MRI) lesion load, spinal cord atrophy [16], and genetics [17]. There is no definitive cure. Corticosteroids may hasten relapse recovery but do not alter long-term outcomes [18]. Since the early 1990s, disease-modifying drugs (DMDs) such as interferon beta (IFN β), glatiramer acetate, natalizumab, fingolimod, alemtuzumab, teriflunomide, and dimethyl fumarate have reduced relapse frequency and severity, but options for progressive MS remain limited, and major unmet needs persist.

Description of the intervention and how it might work

Cladribine is a purine nucleoside analogue that selectively depletes peripheral lymphocytes while largely sparing the innate immune system and modulating CNS-resident microglia [19]. It acts as a prodrug, activated through intracellular triphosphate accumulation. Three key enzymes (deoxycytidine kinase, 5'-nucleotidase, and adenosine deaminase) regulate its activation, leading to interference with cell metabolism, DNA damage, and apoptosis.

Cladribine preferentially targets lymphocytes, producing sustained reductions in CD4+ and CD8+ cells, with transient effects on CD19+ cells, while sparing other immune populations [20, 21]. Its immunological effect persists for over 10 months after the last dose, influencing both resting and dividing lymphocytes [22], which enables short annual treatment courses. Limited effects on innate immunity may preserve overall immunocompetence.

Mechanistically, cladribine is associated with favourable biomarker changes, such as reduced neurofilament light chain (NfL) levels – indicating less neuroaxonal damage – and fewer active MRI lesions [23]. By depleting T and B cells in both the peripheral and central nervous system, it targets key processes in MS pathogenesis.

Cladribine has been used for decades in haematological disorders including hairy cell leukaemia and B-cell chronic lymphocytic leukaemia [24, 25], and it has been investigated in other autoimmune conditions [26].

Why it is important to do this review

Cladribine was first tested in PRMS through four RCTs with parenteral administration in the USA, with encouraging results for slowing disability progression (Sipe 1994 [27, 28]; Rice 2000 [29]; Romine 1999 [30], Stelmasiak 2009 [31]).

More recently, researchers investigated oral cladribine in different MS phenotypes. Pivotal trials such as CLARITY (Giovannoni 2010 [32, 33]), ORACLE (Leist 2014 [34]), and ONWARD (Montalban 2018_amended [35]; Montalban 2018_original protocol [36]) reported favourable results for relapse rates, disability progression, and MRI outcomes.

Oral cladribine was approved for highly active RRMS in Europe in 2017 and in the USA in 2019 [37, 38], following an initial refusal by the European Medicines Agency (EMA) in 2010 due to cancer risk in the CLARITY trial, which also led to early ORACLE termination by the study sponsor [39]. Later data from the CLARITY Extension (Giovannoni 2018 [40, 41]), the PREMIERE register (Leist 2020 [42]), and re-analysis of the CLARITY trial showed no clear increase in malignancy risk [43, 44].

In 2023, the World Health Organization (WHO) added cladribine, along with glatiramer acetate and rituximab, to its essential medicines list for MS [45]. Although cladribine appears effective and well tolerated, its risk-benefit profile across MS subtypes and long-term outcomes remains unclear.

OBJECTIVES

To assess the short- and long-term benefits and harms of cladribine compared to no intervention, placebo, or any other disease-modifying drugs (DMDs) in people with any form of multiple sclerosis (MS).

METHODS

We conducted this review according to the Methodological Expectations of Cochrane Intervention Reviews (MECIR) standards and reported following PRISMA 2020 [46].

Differences between protocol and review

The protocol planned subgroup analyses if more than 10 studies were available. Since we identified nine or fewer eligible studies of

each type, subgroup results were limited. No data were available by age. Four studies provided data for treatment-naïve participants, but we did not analyse this subgroup because the sample sizes were too small. Subgroup comparisons by administration route, dosage, or MS subtype did not alter our main results and were unreliable due to the small number of studies.

Criteria for considering studies for this review

Types of studies

We included randomised controlled trials (RCTs), their open-label extensions (OLEs), the first phase of cross-over trials, and non-randomised studies of interventions (NRSIs). We decided to include OLEs and NRSIs mainly to detect rare adverse events (e.g. malignancies) and long-term disability progression. Eligibility was independent of publication status or language. We excluded retrospective studies and those without a comparator.

Types of participants

We included people with any form of MS or CIS, diagnosed according to accepted international criteria [47, 48, 49], regardless of age, sex, race, disability, and disease duration. In studies with a mix of eligible and ineligible participants, we analysed data from eligible participants only.

Types of interventions

We investigated oral or parenteral cladribine, alone or combined, at any dose or duration, versus placebo, no intervention, or other DMDs. Concomitant interventions were accepted if used across all groups.

Outcome measures

A panel of four neurologists and four randomly selected patients of mixed disability and sex prioritised outcomes using the GRADE 1-to-9 scale (7–9: critical; 4–6: important; 1–3: limited importance; [Supplementary material 8](#)) [50]. We excluded studies that reported no relevant outcomes.

Critical outcomes

We considered the following critical outcomes, measured at the end of follow-up.

- Total number of SAEs. If too few studies reported the total number of SAEs and person-years, we planned to use the number of participants who had experienced at least one SAE by the end of follow-up, as defined in the included studies.
- Number of participants free from disability worsening. Disability worsening was defined as an increase in the Expanded Disability Status Scale (EDSS) of at least 1.0 point from baseline, sustained on subsequent visits for at least 12 weeks if the baseline score was 5.5 or less, or an increase of at least 0.5 points sustained for at least 12 weeks if the baseline score was more than 5.5 [51].
- Number of participants with no evidence of disease activity (NEDA). NEDA is a composite outcome that covers freedom from relapse, freedom from EDSS worsening, and lack of new/enlarging T2 lesions or gadolinium-enhancing lesions on MRI [52, 53, 54]. However, as there is no strictly accepted definition of NEDA to date [55], we accepted definitions used in the included studies.

- Number of participants who withdrew from treatment due to any adverse event, lack of efficacy, lack of compliance, or a combination of these. We used this outcome as a measure of global effectiveness.
- Number of participants with evidence of cognitive impairment, assessed according to validated neurocognitive batteries, such as the Brief Repeatable Battery of Neuropsychological Tests (BRB-N) [56], Minimal Assessment of Cognitive Function in MS (MACFIMS) [57], Brief International Cognitive Assessment for MS (BICAMS) [58], or the composite MS Cognitive Battery (MS COG) [59, 60]. Considering the effects of depression, anxiety, and fatigue on cognition [61], we recorded simultaneous reporting of these conditions only where studies used validated scales (Hospital Anxiety and Depression Scale [62] and Fatigue Severity Scale [63]).
- Quality of life (QoL), measured with validated health-related scales (e.g. the Medical Outcomes Study 36-item Short Form Health Survey (SF-36) [64]) or MS-specific scales (MS Impact Scale-29 [65], Leeds MS QoL [66], MS QoL-54 [67], or any other validated scale).

Important outcomes

We considered the following important outcomes, measured at the end of follow-up.

- Number of participants with at least one adverse event (describing type and severity).
- Number of participants with at least one new relapse, defined as new/worsened symptoms lasting more than 24 hours, not due to fever, and preceded by at least one month of stability [47].
- Number of participants with new gadolinium-enhancing T1-weighted (Gd+ T1) MRI lesions or new/enlarging T2-weighted MRI lesions.

Outcomes of limited importance

- Number of participants with evidence of brain atrophy as defined in the included studies, at 24 months and at the end of follow-up, compared with baseline.

Search methods for identification of studies

The complete search strategy is available in [Supplementary material 1](#).

Electronic searches

The Information Specialist searched the Cochrane MS and Rare Diseases of the CNS System's Trials Register (<https://ms.cochrane.org/>) up to 3 February 2025. This register includes records from the following sources.

- Cochrane Central Register of Controlled Trials (CENTRAL; Issue 1, 2025) in the Cochrane Library.
- MEDLINE (PubMed; 1966 to 3 February 2025).
- Embase (Embase.com; 1974 to 3 February 2025).
- CINAHL (EBSCOhost; 1981 to 3 February 2025).
- Latin American and Caribbean Health Science Information Database (LILACS; Bireme; 1982 to 3 February 2025).
- WHO International Clinical Trial Registry Platform (ICTRP; apps.who.int/trialsearch).

- US National Institutes of Health Ongoing Trials Register ClinicalTrials.gov (<https://clinicaltrials.gov/>).

Searching other resources

We used the following additional methods.

- Checked reference lists of relevant reviews and included articles.
- Contacted trial authors and experts for missing or unpublished data.
- Searched conference reports on neurology, MS societies (<https://www.nationalmssociety.org/>), and congresses organised by the European Committee for Treatment and Research in MS (<https://2022.ectrims-congress.eu/>), and by the American Committee for Treatment and Research in MS (<https://www.actrims.org/>).
- Contacted biotechnology companies for published or unpublished studies.
- Searched for postpublication amendments or corrections to eligible studies, by reviewing prior Cochrane reviews, PubMed/Scopus/Web of Science, journal websites, CrossMark, ORCID, and trial registries; maintained alerts and dated logs of findings.

Data collection and analysis

Selection of studies

Three review authors (MM, MO, MVE) independently screened titles and abstracts, retrieving full texts to assess eligibility. We linked multiple publications as companion reports and excluded true duplicates. Disagreements were resolved through weekly online discussions with all review authors.

Data extraction and management

Three review authors (MM, MO, MVE) independently extracted the following study data using standardised forms and entered them into RevMan [68].

- Publication details (date, country, journal, authors).
- Study design, setting, eligibility criteria, and participant characteristics.
- Intervention dose, frequency, scheme, and duration.
- Critical, important, and 'of limited importance' outcomes.
- Follow-up duration and trial period.
- Declarations of interest and funding.

Risk of bias assessment in included studies

Three review authors (MGC, TAC, MO) independently assessed risk of bias and methodological quality, resolving disagreements by discussion. As studies had different designs, we applied design-specific risk of bias tools.

Randomised controlled trials and open-label extension trials

For assessing the risk of bias in RCTs and OLEs, we used the Cochrane RoB 2 tool, which covers the following five domains [69].

- Bias arising from the randomisation process.
- Bias due to deviations from the intended interventions (effects of assignment and adherence to intervention).
- Bias due to missing outcome data.
- Bias in measurement of the outcome.

- Bias in the selection of the reported result.

Each trial received a risk of bias judgement for each outcome (low risk of bias, some concerns, or high risk of bias), based on domain-level judgements, as follows.

- Low risk of bias: if we rated the study at low risk of bias for all domains for the considered outcome.
- Some concerns: if we judged the study to raise some concerns in at least one domain for the considered outcome, but not to be at high risk of bias for any domain
- High risk of bias: if we judged the study to be at high risk of bias in at least one domain for the considered outcome, or if we judged the study to have some concerns for multiple domains in a way that substantially lowered our confidence in the result.

Judgements were supported with explicit evidence, managed in the RoB 2 Excel tool (accessed 19 January 2024) [70]. To implement the RoB 2 tool, we used RevMan [68].

We estimated effects both by intention to treat (assignment at baseline) and per protocol (adherence) for SAEs and adverse events. For cross-over trials, we assessed only the first phases [69].

Non-randomised studies of interventions

We assessed risk of bias in NRSIs using the Cochrane ROBINS-I tool, which covers confounding, participant selection, intervention classification, deviations from intended interventions, missing data, outcome measurement, and selection of reported results [71].

We assigned judgements for each domain (low risk, moderate risk, serious risk, critical risk, or no information) based on the answers to signalling questions (yes, probably yes, no, probably no, or no information). We then assigned an overall risk of bias judgement for each outcome, as follows.

- Low: comparable to a well-conducted RCT.
- Moderate: sound for an NRSI but not comparable to an RCT.
- Serious: important problems in at least one domain.
- Critical: too problematic for reliable evidence.
- No information: insufficient data in key domains.

We prespecified confounders likely to influence treatment effects (age, sex, MS type, and disease duration). For each NRSI, we noted whether these were controlled and whether appropriate statistical methods were used. We identified no relevant cointerventions.

Measures of treatment effect

Data were analysed in RevMan [68]. For dichotomous outcomes, we reported risk ratios (RRs) with 95% confidence intervals (CIs). We planned to report QoL using mean differences (MDs) with 95% CIs.

Unit of analysis issues

The unit of analysis was the individual participant. For cross-over studies, we used only first-phase data to avoid carry-over effects or bias from disease evolution.

We treated multiarm trials as independent two-arm comparisons. When studies compared different cladribine doses with the same control, we merged doses and summed events and sample sizes to prevent double counting and correlation of results.

Dealing with missing data

We contacted study investigators and sponsors to request missing data or clarify inconsistencies. Although our protocol planned sensitivity analyses for losses to follow-up or exclusions [72], we ultimately judged these would not add useful information beyond the most plausible scenario described in the *Cochrane Handbook for Systematic Reviews of Interventions* [73].

Reporting bias assessment

As each meta-analysis included fewer than 10 studies, we did not create funnel plots to explore small-study effects or publication bias.

Synthesis methods

We pooled clinically comparable studies in meta-analyses when methods and data were sufficiently similar. Separate analyses were conducted for RCTs, OLEs, and NRSIs, using the inverse-variance random-effects model to account for differences in methodological quality, population severity, and intervention efficacy.

Investigation of heterogeneity and subgroup analysis

We assessed clinical and methodological heterogeneity by comparing participant and intervention characteristics, and statistical heterogeneity with the χ^2 test ($p < 0.1$) and I^2 statistic [74]. Where we identified substantial heterogeneity ($I^2 > 50\%$), we examined its potential sources [75].

When at least 10 studies were available for analysis, we planned to explore the following subgroups.

- Treatment-naïve versus previously treated participants.
- RRMS versus CIS versus PPMS versus SPMS.
- Age below 18 years versus 18 years or older.
- Cladribine dosage: high (5.25 mg/kg cumulative dose) versus low (2.1–3.5 mg/kg cumulative dose).
- Administration route: oral versus intravenous versus subcutaneous.

Equity-related assessment

We considered potential inequities in access to cladribine or effects of cladribine among disadvantaged groups (e.g. by sex, socioeconomic status, or geography). We planned to extract and report equity-relevant data (ethnicity, socioeconomic status, location).

Sensitivity analysis

We did not reanalyse data excluding studies at high risk of bias, as nearly all included studies were judged at high risk.

Certainty of the evidence assessment

We assessed the certainty of evidence for critical and important outcomes using the GRADE approach, considering internal validity (risk of bias, inconsistency, imprecision, publication bias) and external validity (directness). We used methods and recommendations presented in the *Cochrane Handbook for Systematic Reviews of Interventions* [76].

We created the following five summary of findings tables.

- Cladribine versus placebo: critical outcomes in RCTs ([Summary of findings 1](#)).
- Cladribine versus placebo: critical outcomes in OLEs ([Summary of findings 2](#)).
- Cladribine versus placebo: important outcomes in RCTs ([Summary of findings 3](#)).
- Cladribine versus placebo: important outcomes in OLEs ([Summary of findings 4](#)).
- Cladribine versus other DMDs: at least one relapse in NRSIs ([Summary of findings 5](#)).

For summary of findings tables 1 and 2, we included the following critical outcomes.

- Number of participants with at least one SAE.
- Number of participants free from disability worsening.
- Number of participants with NEDA.
- Number of participants who withdrew from treatment.
- Number of participants with evidence of cognitive impairment.
- QoL.

For summary of findings tables 3 and 4, we included the following important outcomes.

- Number of participants with at least one adverse event.
- Number of participants with at least one relapse.
- Number of participants with new Gd+ T1 MRI lesions.

Consumer involvement

Consumers participated in ranking MS outcome measures, ensuring inclusion of patient perspectives and values ([Supplementary material 8](#)). A caregiver wrote the plain language summary.

RESULTS

Description of studies

Full study descriptions and PICO questions are available in [Supplementary material 2](#), [Supplementary material 3](#), and [Supplementary material 4](#). No studies were awaiting classification. [Supplementary material 6](#) provides a complete record of all analyses, and [Supplementary material 7](#) provides a full data package. We contacted authors or sponsors of six studies for additional data [77, 78, 79, 80, 81, 82, 83, 84, 85], and we obtained further information for four.

Results of the search

The search (3 February 2025) identified 1824 records. After removing 508 duplicates, we screened the titles and abstracts of 1316 records, excluding 1261 as clearly irrelevant. We then retrieved and assessed 55 full-text articles, excluding 32. We included 15 studies reported in 18 articles. Nine studies were RCTs (Giovannoni 2010; Janiec 2001 [86]; Leist 2014; Montalban 2018_amended; Montalban 2018_original protocol; Rice 2000; Romine 1999; Sipe 1994; Stelmasiak 2009), four were OLEs (Giovannoni 2018; Giovannoni 2023 [87]; Giovannoni 2025 [88]; Leist 2014 _OLMP [89]), and two were NRSIs (Kalincik 2018 [90]; Spelman 2023 [91]). There were three reports relating to MRI outcomes [27, 32, 40]; see [Table 1](#) for details. We identified

five potentially eligible ongoing studies (ChariotMS ongoing [92]; CLARION ongoing [93]; CLASP-MS [94]; Hellwig 2025 [95]; NOR-MS ongoing [96]). [Figure 1](#) presents the study selection process in a PRISMA diagram.

Figure 1. PRISMA flow diagram. Search updated to 3 February 2025.

OLE: open-label extension; NRSI: nonrandomised study of intervention; RCT: randomized controlled trial.

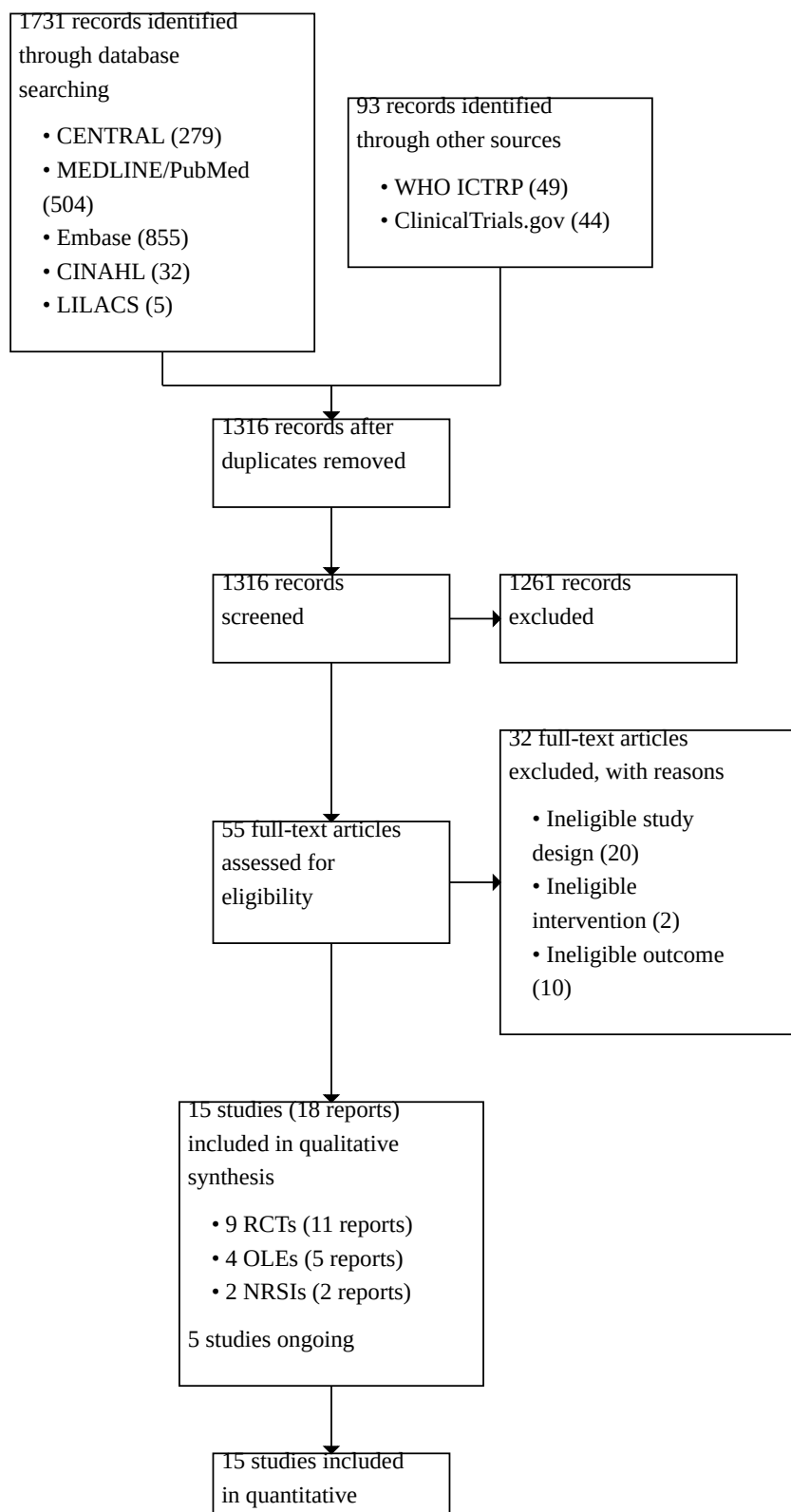


Figure 1. (Continued)

10 studies included
in quantitative
synthesis
(meta-analysis)

Included studies

Randomised controlled trials

We included nine RCTs (2571 participants) published between 1996 and 2018.

Study design and geographical setting

Four RCTs were multicentre, double-blind, placebo-controlled, parallel-group trials: Giovannoni 2010 (155 centres/32 countries), Leist 2014 (160 centres/34 countries), Montalban 2018_original protocol (50 centres/four countries) and Rice 2000 (six centres/two countries). Romine 1999 was a single-centre parallel-group trial (Scripps Clinic, USA). Three RCTs were single-centre, double-blind, cross-over studies: Janiec 2001 and Stelmasiak 2009 were conducted in Poland, Sipe 1994 in the USA.

Participants

Participants were adults aged 18 to 65 years (mean 39.9 years) with MS of varying severity (mean EDSS 3.5), ranging from the most severe participants in Rice 2000 (mean EDSS 5.6) to those in the best condition in Leist 2014 (mean EDSS 1.2).

Three RCTs included 1642 participants with RRMS (Giovannoni 2010; Romine 1999; Stelmasiak 2009). All participants had active disease (≥ 1 T1 Gd+ MRI lesion). Two RCTs (210 participants) included people with progressive MS: Sipe 1994 used Poser Criteria (defined as chronic progressive for signs and symptoms over the preceding 24 months), while Rice 2000 used Poser or Schumacher criteria (over the preceding 12 months). Three RCTs included people with mixed MS forms: Janiec 2001 included 69 participants with PRMS and SPMS forms at the active stage of the disease, while Montalban 2018_original protocol and Montalban 2018_amended included 214 participants with RRMS or SPMS. Just one RCT included people with CIS (Leist 2014; 616 participants).

All studies excluded people at high infection risk or with immunosuppression. Stelmasiak 2009 included treatment-naïve people. Only Leist 2014 allowed prior corticosteroid use. All other studies excluded people treated with corticosteroids and other immunosuppressive drugs during varying time periods (one to nine months) before randomisation. Giovannoni 2010 and Montalban 2018_original protocol also excluded people with recent (within three months) cytokine-based therapy, intravenous immunoglobulin, or plasmapheresis.

Interventions

Giovannoni 2010, Leist 2014, and Montalban 2018_original protocol randomised oral cladribine in one of two cumulative doses (3.5 mg/kg or 5.25 mg/kg) or matching placebo. In Montalban 2018_original protocol, 5.25 mg/kg was discontinued following adverse reports in

88.2% of participants. We analysed the amended study separately (Montalban 2018_amended).

Three studies used daily subcutaneous injections, with consecutive cladribine or placebo; the cumulative dose of cladribine was 2.1 mg/kg in Janiec 2001 and Romine 1999, and 2.45 mg/kg in Stelmasiak 2009. Rice 2000 evaluated subcutaneous cladribine at a cumulative dose of 0.7 mg/kg versus subcutaneous cladribine at a cumulative dose of 0.7 mg/kg versus placebo. In each study, cycles were repeated monthly up to the quantity of cumulative doses.

Sipe 1994 administered cladribine intravenously with a cumulative dose of 2.8 mg/kg in monthly courses, comparing with placebo.

Outcome measures

Outcome timing was quite short, ranging from 52 weeks (Sipe 1994; Stelmasiak 2009; Rice 2000) to 104 weeks (Janiec 2001). Three studies lasted 96 weeks (Giovannoni 2010; Leist 2014; Montalban 2018_amended).

As primary endpoints, Romine 1999 and Giovannoni 2010 assessed relapse rate, Leist 2014 chose time-to-conversion to clinically definite MS, Montalban 2018_original protocol considered safety and tolerability, three studies assessed disability (Rice 2000; Sipe 1994; Stelmasiak 2009), and Janiec 2001 examined leukocyte levels. All studies except Janiec 2001 considered the number of participants with SAEs. Only three trials provided data on disability measured with EDSS per participant (Giovannoni 2010; Montalban 2018_amended; Sipe 1994); the others reported mean differences in EDSS. All studies except Janiec 2001 considered the number of participants who withdrew from treatment, and all except Rice 2000 reported the number of participants with at least one adverse event (Rice 2000 reported number of adverse events). Only two trials considered the number of participants with at least one relapse by the end of follow-up (Giovannoni 2010; Montalban 2018_amended). New Gd+ T1 lesions (per participant) were recurrent MRI endpoints, found in only four trials (Giovannoni 2010; Leist 2014; Rice 2000; Sipe 1994). Giovannoni 2010 and Sipe 1994 offered a new publication of MRI endpoints, where this outcome was well specified. Leist 2014 and Montalban 2018_original protocol considered the number of MRI lesions per scan: Gd+ T1 lesions, active T2-weighted lesions, and combined unique lesions. In Montalban 2018_amended, we found only the mean number of new Gd+ T1 and active T2-weighted lesions per participant per scan. In Romine 1999, a second primary endpoint was the number of Gd+ T1 lesions, but we were unable to extract data from the short text. Similarly, in Stelmasiak 2009, the endpoint was stated but not reported.

No RCTs reported NEDA, QoL, or cognitive impairment. Brain atrophy in MRI data were inconsistent across studies.

Funding

Four studies received funding or support from pharmaceutical companies (Giovannoni 2010; Leist 2014; Montalban 2018_amended; Montalban 2018_original protocol). Rice 2000 did not provide this information, Romine 1999 had mixed public-private funding, and the remaining RCTs were funded by governmental institutions or private philanthropic foundations (Janiec 2001; Sipe 1994; Stelmasiak 2009).

Open-label extension trials

We included four OLEs: Giovannoni 2018 and Giovannoni 2023 were derived from the CLARITY trial (Giovannoni 2010), while Leist 2014 _OLMP and Giovannoni 2025 were derived from the ORACLE trial (Leist 2014).

Study design and geographical setting

In the CLARITY extension trial, placebo recipients switched to cladribine 3.5 mg/kg, while cladribine-treated participants were re-randomised (2:1) to cladribine 3.5 mg/kg or placebo, allowing assessment of two extra years of therapy. In ORACLE, participants were grouped by conversion to clinically defined MS (CDMS), entering the open-label maintenance period (OLMP) or long-term follow-up (planned 96 weeks, ended early by sponsor).

Population

In Giovannoni 2018 (not a preplanned study), participants entered the extension a median of 40.3 weeks after CLARITY termination. Of 1326 participants originally randomised, only 806 (60.8%) were re-enrolled. The mean age of participants was 38.5 years, 65.9% were women, and the mean EDSS score was 2.6 (range 2 to 6). In Leist 2014 _OLMP, 109 of 616 participants included in Leist 2014 converted to CDMS (mean 49.9 weeks) and entered the OLMP (mean age 31 years, 63.5% women). Giovannoni 2023 included 435 participants from CLARITY (345 also in the extension). Giovannoni 2025 analysed 227 of 616 original ORACLE-MS participants.

Intervention

Each study evaluated oral cladribine. In Giovannoni 2018, treatment comprised one or two 10-mg tablets daily for four or five days per course. Across CLARITY and its extension, cumulative cladribine doses ranged from 3.5 mg/kg to 8.75 mg/kg; in the extension, placebo was administered to participants previously treated with cladribine 3.5 mg/kg or 5.25 mg/kg. In Leist 2014 _OLMP, converters received INT- β 44 μ g three times weekly.

Outcome

Follow-up lasted 96 weeks in Giovannoni 2018 and Leist 2014 _OLMP, 10.9 years in Giovannoni 2023, and 13.1 years in Giovannoni 2025. Two OLEs reported SAEs and withdrawals (Giovannoni 2018, Leist 2014 _OLMP), only Giovannoni 2018 reported disability and new Gd+ T1 MRI lesions, three OLEs reported new relapse (Giovannoni 2018; Giovannoni 2023; Giovannoni 2025), and two reported NEDA (Giovannoni 2023; Giovannoni 2025).

Funding

All the studies were sponsored by pharmaceutical companies.

Non-randomised studies of interventions

We included two propensity score-matched observational studies (Kalincik 2018; Spelman 2023). Both used the MSBase registry,

an international observational cohort [97]. In Kalincik 2018, 111 people received cladribine (90 from the Australian Product Familiarisation Program, 21 off label); of these, 37 people were eligible for matching. In Spelman 2023, 633 people were treated with oral cladribine. Data were collected during routine clinical visits in tertiary MS centres. Kalincik 2018 compared cladribine with IFN- β , fingolimod, and natalizumab; Spelman 2023 compared cladribine with fingolimod, dimethyl fumarate, and teriflunomide. Both studies assessed the number of participants with at least one new relapse at the end of follow-up (one year in Kalincik 2018; three years in Spelman 2023).

Funding

Kalincik 2018 had mixed public-private funding, while Spelman 2023 was sponsored by pharmaceutical companies.

Excluded studies

[Supplementary material 3](#) provides reasons for the exclusion of 32 studies.

- Ineligible design (20 studies): one case series, one without a control group, six without original data, two with pooled populations, four post-hoc analyses, and six ongoing single-arm studies.
- Ineligible interventions (two studies).
- Irrelevant outcomes (10 studies): MRI outcomes not aligned with our protocol in eight studies, non-EDSS disability scale in one, and events (not participants) counted in one.

Risk of bias in included studies

Risk of bias assessments for each outcome, with domain-level judgements and rationale, are shown beside each forest plot and summarised in [Supplementary material 5](#).

Randomised controlled trials

We rated no RCTs at overall low risk of bias. Across outcomes, most showed some concerns or a high risk of bias, mainly due to baseline imbalances, unclear allocation concealment, and absence of prespecified analysis plans.

Overall risk of bias by outcome

- At least one SAE at the end of follow-up (eight studies)
 - High risk of bias: six studies (Leist 2014; Montalban 2018_amended; Rice 2000; Romine 1999; Sipe 1994; Stelmasiak 2009)
 - Some concerns: two studies (Giovannoni 2010; Montalban 2018_original protocol)
 - Main issues: randomisation, intervention deviations
 - Least biased: missing outcome data (except Sipe 1994)
- At least one adverse event at the end of follow-up (eight studies)
 - High risk of bias: six studies (Janiec 2001; Leist 2014; Montalban 2018_amended; Romine 1999; Sipe 1994; Stelmasiak 2009)
 - Some concerns: two studies (Giovannoni 2010; Montalban 2018_original protocol)
 - Main issues: intervention deviations, outcome measurement
- Treatment withdrawal before the end of follow-up (eight studies)

- High risk of bias: five studies (Leist 2014; Montalban 2018_amended; Rice 2000; Sipe 1994; Stelmasiak 2009)
- Some concerns: three studies (Giovannoni 2010; Montalban 2018_original protocol; Romine 1999)
- Main issues: intervention, deviations, measurement
- Freedom from disability worsening at the end of follow-up (three studies)
 - High risk of bias: two studies (Montalban 2018_amended and Sipe 1994)
 - Some concerns: one study (Giovannoni 2010)
 - Main issues: randomisation and intervention deviations
- At least one new relapse by the end of follow-up (two studies)
 - High risk of bias: one study (Montalban 2018_amended)
 - Some concerns: one study (Giovannoni 2010)
 - Main concerns: intervention deviations, selective reporting
- New Gd+ T1 MRI lesions at the end of follow-up (four studies)
 - High risk of bias: three studies (Sipe 1994, Leist 2014; Rice 2000)
 - Some concerns: one study (Giovannoni 2010)

- Main issues: randomisation, intervention deviations

Open-label extension trials

All four OLEs were rated at high overall risk of bias across outcomes (Giovannoni 2018; Giovannoni 2023; Giovannoni 2025; Leist 2014 _OLMP). Bias mainly arose from randomisation, intervention deviations, and missing outcome data. Outcome measurement was generally low risk, except for withdrawals, where reasons were often unclear ('other'). Selective reporting was low risk only for NEDA; for other outcomes, some concerns remained, as two earlier OLEs lacked prespecified analysis plans (Giovannoni 2018; Leist 2014 _OLMP), and the two later OLEs analysed annualised relapse rate rather than the number of participants with relapses (Giovannoni 2023; Giovannoni 2025).

Non-randomised studies of interventions

Using ROBINS-I, we rated Kalincik 2018 and Spelman 2023 at serious overall risk, mainly from participant selection and outcome measurement. Spelman 2023 also had high risk of bias in the confounding and missing data domains (Table 2; Table 3; Figure 2).

Figure 2. Comparison: Cladribine versus other DMDs for multiple sclerosis (MS) – results from NRSIs - Outcome: Number of participants who experienced at least one new relapse at the end of follow-up

		Risk of bias domains							
		D1	D2	D3	D4	D5	D6	D7	Overall
Study	Kalincik 2018								
	Spelman 2023								
Domains:									
D1: Bias due to confounding.									
D2: Bias due to selection of participants.									
D3: Bias in classification of interventions.									
D4: Bias due to deviations from intended interventions.									
D5: Bias due to missing data.									
D6: Bias in measurement of outcomes.									
D7: Bias in selection of the reported result.									
Judgement									
 Serious									
 Moderate									
 Low									
 No information									

Synthesis of results

Cladribine compared to placebo

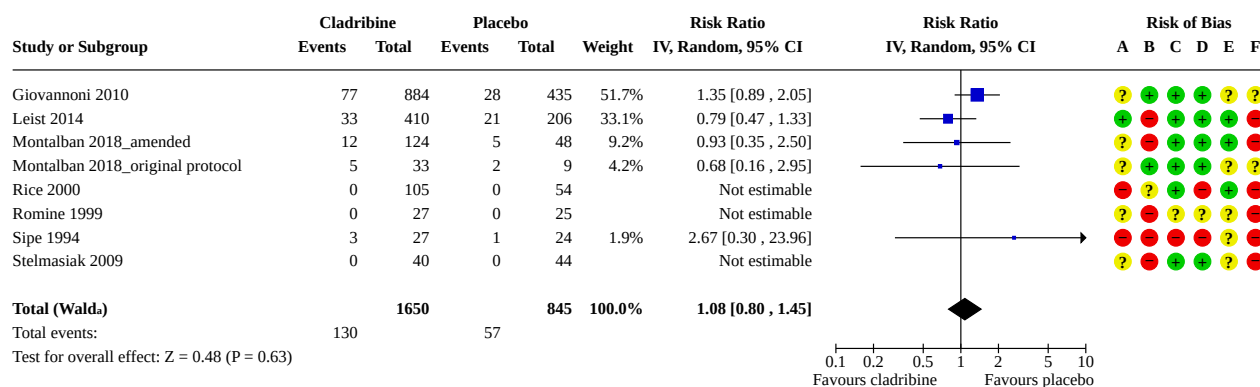
Results from randomised controlled trials

Critical outcomes

Summary of findings 1 summarises the effect estimates for cladribine. Only three of six critical outcomes were available.

Number of participants with at least one serious adverse event

Cladribine versus placebo may have little to no effect on the risk of experiencing at least one SAE over 48 to 96 weeks in a population adhering to the intervention (per-protocol effect), but the evidence is very uncertain (RR 1.08, 95% CI 0.80 to 1.45; $I^2 = 0\%$; 8 studies, 2495 participants; very low-certainty evidence; Figure 3). We downgraded the evidence certainty for very serious risk of bias and serious imprecision.

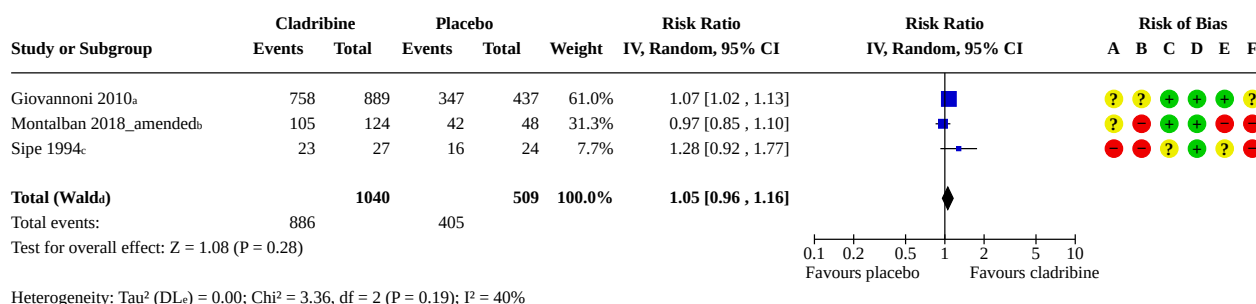
Figure 3. Analysis 1.1**Comparison: cladribine versus placebo for multiple sclerosis (MS) – results from RCTs - Outcome: N° of participants with at least one serious adverse event (12- to 22-month follow-up)****Footnotes**^aCI calculated by Wald-type method.^bTau² calculated by DerSimonian and Laird method.**Risk of bias legend**

- (A) Bias arising from the randomization process
(B) Bias due to deviations from intended interventions
(C) Bias due to missing outcome data
(D) Bias in measurement of the outcome
(E) Bias in selection of the reported result
(F) Overall bias

Number of participants free from disability worsening

Cladribine versus placebo may result in little to no difference in the number of participants who are free from disability worsening up to 24 months (RR 1.05, 95% CI 0.96 to 1.16; I² = 40%; 3 studies, 1549

participants; low-certainty evidence; [Figure 4](#)). This represents 835 people free from disability of 1000 people treated with cladribine, versus 796 people free from disability of 1000 treated with placebo. We downgraded the evidence certainty for very serious risk of bias in two trials.

Figure 4. Analysis 1.2**Comparison: cladribine versus placebo for multiple sclerosis (MS) – results from RCTs - Outcome: N° of participants free from the EDSS worsening (22-month follow-up)****Footnotes**^aat the end of 96 weeks FU (approximated to 24months)^bat 24 months, Amended Protocol^cat 12 months^dCI calculated by Wald-type method.^eTau² calculated by DerSimonian and Laird method.**Risk of bias legend**

- (A) Bias arising from the randomization process
- (B) Bias due to deviations from intended interventions
- (C) Bias due to missing outcome data
- (D) Bias in measurement of the outcome
- (E) Bias in selection of the reported result
- (F) Overall bias

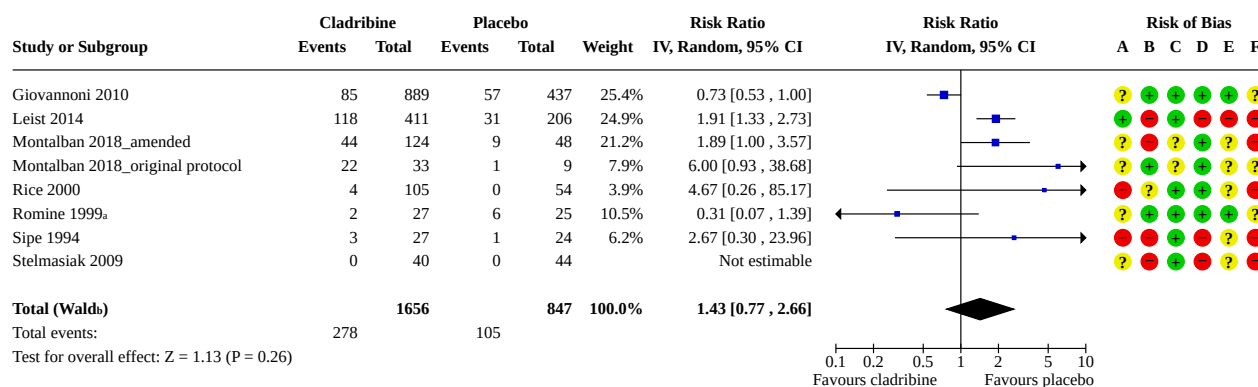
Number of participants with no evidence of disease activity

No RCTs reported NEDA.

Number of participants who withdrew from treatment

Cladribine versus placebo may have little to no effect on the risk of treatment withdrawal from 48 to 96 weeks, but the evidence

is very uncertain (RR 1.43, 95% CI 0.77 to 2.66; I² = 76%; 8 studies, 2503 participants; very low-certainty evidence; [Figure 5](#)). We downgraded the evidence certainty for very serious risk of bias, serious imprecision, and serious inconsistency. There was substantial heterogeneity.

Figure 5. Analysis 1.3**Comparison: cladribine versus placebo for multiple sclerosis (MS) – results from RCTs - Outcome: N° of participants who withdrew (12- to 22-month follow-up)****Footnotes**^adata at 18 months^bCI calculated by Wald-type method.^cTau² calculated by DerSimonian and Laird method.**Risk of bias legend**

- (A) Bias arising from the randomization process
- (B) Bias due to deviations from intended interventions
- (C) Bias due to missing outcome data
- (D) Bias in measurement of the outcome
- (E) Bias in selection of the reported result
- (F) Overall bias

Number of participants with evidence of cognitive impairment

No RCTs reported evidence of cognitive impairment.

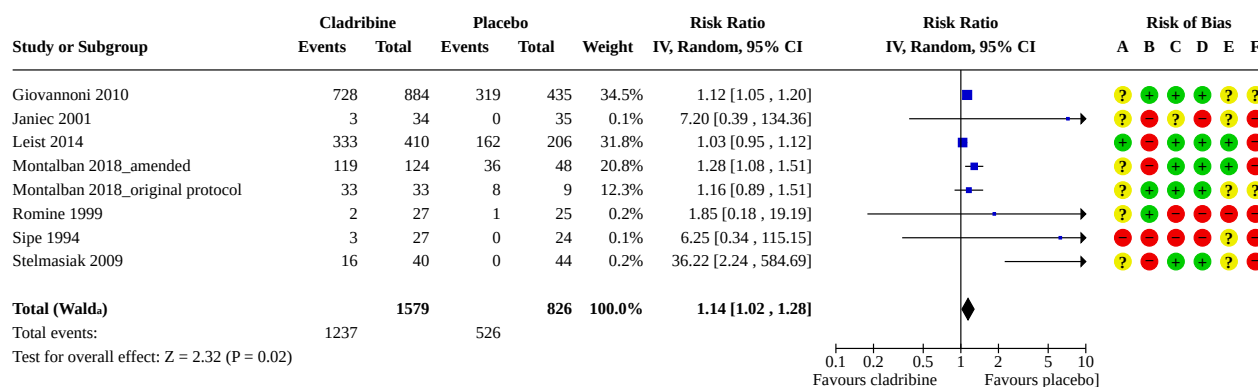
Quality of life

No RCTs reported QoL.

Important outcomesSee [Summary of findings 3](#).**Number of participants with at least one adverse event**

Cladribine versus placebo may increase the risk of experiencing at least one adverse event by the end of follow-up for a population

adhering to the intervention (per-protocol effect), but the evidence is very uncertain (RR 1.14, 95% CI 1.02 to 1.28; I² = 53%; 8 studies, 2405 participants; very low-certainty evidence; [Figure 6](#)). All trials except Rice 2000 reported information on adverse events. We downgraded the evidence certainty for very serious risk of bias and serious inconsistency. In the subgroup analysis, the test for subgroup differences suggested a subgroup effect (P = 0.06) according to the MS clinical form; there were more adverse events in participants with RRMS/PRMS (RR 1.19, 95% CI 1.02 to 1.39; underpowered).

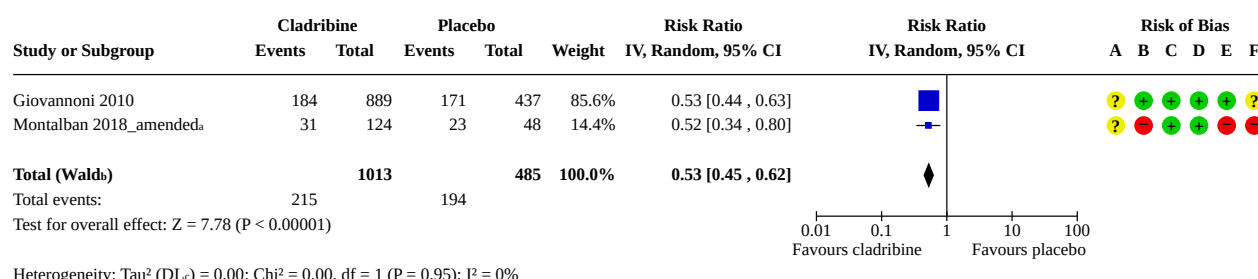
Figure 6. Analysis 1.4**Comparison: cladribine versus placebo for multiple sclerosis (MS) – results from RCTs - Outcome: N° of participants with at least one adverse event (12- to 24-month follow-up)****Footnotes**^aCI calculated by Wald-type method.^bTau² calculated by DerSimonian and Laird method.**Risk of bias legend**

- (A) Bias arising from the randomization process
(B) Bias due to deviations from intended interventions
(C) Bias due to missing outcome data
(D) Bias in measurement of the outcome
(E) Bias in selection of the reported result
(F) Overall bias

Number of participants with at least one new relapse

Cladribine versus placebo likely reduces the risk of experiencing at least one new relapse by the end of follow-up (RR 0.53, 95% CI 0.45 to 0.62; I² = 0%; 2 studies, 1498 participants; moderate-certainty

evidence; Figure 7). This represents 212 people experiencing a new relapse by 96 weeks out of 1000 people receiving cladribine, compared with 400 people experiencing a new relapse out of 1000 receiving placebo. We downgraded the evidence certainty for serious risk of bias in one study.

Figure 7. Analysis 1.5**Comparison: cladribine versus placebo for multiple sclerosis (MS) – results from RCTs - Outcome: N° of participants with at least one relapse (22-month follow-up)****Footnotes**^adata from clinicaltrial.gov^bCI calculated by Wald-type method.^cTau² calculated by DerSimonian and Laird method.**Risk of bias legend**

- (A) Bias arising from the randomization process
(B) Bias due to deviations from intended interventions
(C) Bias due to missing outcome data
(D) Bias in measurement of the outcome
(E) Bias in selection of the reported result
(F) Overall bias

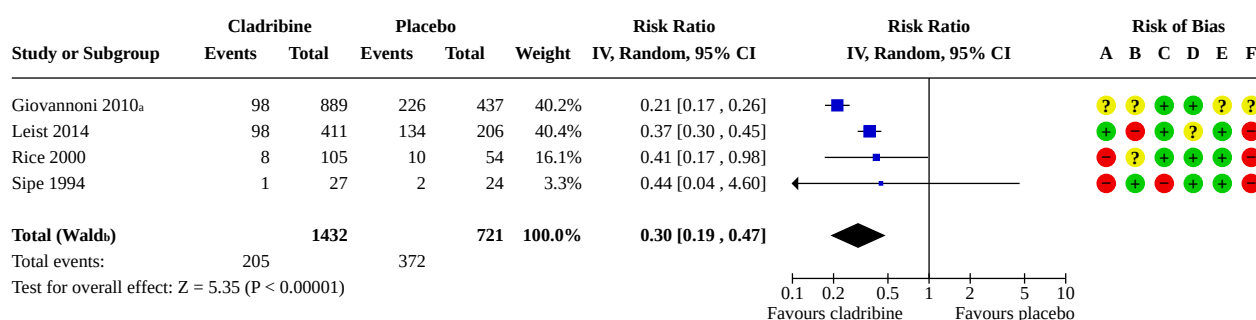
Number of participants with new gadolinium-enhancing T1-weighted MRI lesions

Cladribine versus placebo may reduce the risk of developing new Gd+ T1 MRI lesions by the end of follow-up, but the evidence is very uncertain (RR 0.30, 95% CI 0.19 to 0.47; $I^2 = 73\%$; 4 studies, 2153 participants; very low-certainty evidence; [Figure 8](#)). This means that for every 1000 people treated with cladribine, 153 will develop new Gd+ T1 MRI lesions, compared with 495 of every 1000

people receiving placebo. We downgraded the evidence certainty for very serious risk of bias and serious inconsistency. There was substantial heterogeneity. All subgroups, though underpowered, favoured cladribine. In the subgroup analysis, the test for subgroup differences indicated a difference according to MS clinical form ($P < 0.001$). The smallest reduction was observed in people with SPMS/PPMS (RR 0.42, 95% CI 0.18 to 0.94) with a wide CI due to the small sample size.

Figure 8. Analysis 1.6

Comparison: cladribine versus placebo for multiple sclerosis (MS) – results from RCTs - Outcome: N° of participants with new Gd+ T1 MRI lesions (12- to 22-month follow-up)

**Footnotes**

^aMRI data from Giovannoni 2010 were published in Comi 2013

^bCI calculated by Wald-type method.

^c Tau^2 calculated by DerSimonian and Laird method.

Risk of bias legend

- (A) Bias arising from the randomization process
- (B) Bias due to deviations from intended interventions
- (C) Bias due to missing outcome data
- (D) Bias in measurement of the outcome
- (E) Bias in selection of the reported result
- (F) Overall bias

Outcomes of limited importance**Number of participants with evidence of brain atrophy**

No RCTs reported evidence of brain atrophy. We excluded studies that reported changes in brain volume.

Results from open-label extension studies

[Summary of findings 2](#) summarises the effect estimates of cladribine. Four OLEs evaluated outcomes beyond two years.

Critical outcomes**Number of participants with at least one serious adverse event**

Cladribine versus placebo may have little to no effect on the risk of experiencing at least one SAE in a population adhering to the intervention (per-protocol effect), but the evidence is very uncertain (RR 0.89, 95% CI 0.60 to 1.33; $I^2 = 0\%$; 2 studies, 915 participants; very low-certainty evidence; Analysis 10.1). We downgraded the evidence certainty for very serious risk of bias, serious imprecision, and serious indirectness. In Giovannoni 2018, 1.4% of participants developed a malignancy: eight of 616 participants who received any dosage of cladribine compared to

three of 190 participants who received placebo (RR 0.82, 95% CI 0.22 to 3.07). Leist 2014 _OLMP also assessed malignancies, but no malignancy events occurred during the study.

Number of participants free from disability worsening

Cladribine versus placebo may have little to no effect on the number of people remaining free from disability worsening measured on the EDSS by the end of follow-up, but the evidence is very uncertain (RR 1.02, 95% CI 0.93 to 1.11; 1 study, 806 participants; very low-certainty evidence; Analysis 10.2). This means that for every 1000 people receiving cladribine, 768 will be free from disability worsening, compared with 753 of 1000 people receiving placebo. We downgraded the evidence certainty for very serious risk of bias and serious imprecision.

Number of participants with no evidence of disease activity

Cladribine versus placebo may increase the number of people with NEDA at four years after randomisation (RR 1.63, 95% CI 1.23 to 2.16; $I^2 = 0\%$; 2 studies, 662 participants; low-certainty evidence; Analysis 10.7). This means that for every 1000 people receiving cladribine, 538 will have NEDA, compared with 330 of 1000 people receiving

placebo. We downgraded the evidence certainty for very serious risk of bias in both studies.

Number of participants who withdrew from treatment

Cladribine versus placebo may have little to no effect on the risk of treatment withdrawal by the end of follow-up, but the evidence is very uncertain (RR 0.93, 95% CI 0.81 to 1.08; $I^2 = 0\%$; 2 studies, 915 participants; very low-certainty evidence; Analysis 10.3). We downgraded the evidence certainty for very serious risk of bias, very serious indirectness, and serious imprecision.

Number of participants with evidence of cognitive impairment

No OLEs reported evidence of cognitive impairment.

Quality of life

No OLEs reported QoL.

Important outcomes

See [Summary of findings 4](#).

Number of participants with at least one adverse event

Cladribine versus placebo may have little to no effect on the risk of experiencing at least one adverse event by the end of follow-up, but the evidence is very uncertain (RR 1.04, 95% CI 0.96 to 1.12; $I^2 = 0\%$; 2 studies, 915 participants; very low-certainty evidence; Analysis 10.4). We downgraded the evidence certainty for very serious risk of bias and very serious indirectness.

Number of participants with at least one new relapse

Cladribine versus placebo may reduce the risk of experiencing at least one new relapse over 120 weeks, but the evidence is very uncertain (RR 0.86, 95% CI 0.67 to 1.10; 1 study, 806 participants; very low-certainty evidence; Analysis 10.5). This means that for every 1000 people receiving cladribine, 276 will experience a new relapse, compared with 321 of 1000 people receiving placebo. Two studies reported this outcome at long-term follow-up (9.5–11.4 years) with similar results (RR 0.68, 95% CI 0.59 to 0.79; 2 studies, 662 participants; very low-certainty evidence; Analysis 10.5). This means that for every 1000 people receiving cladribine, 492 will experience a new relapse, compared with 723 of 1000 people receiving placebo. We downgraded the evidence certainty in both analyses for very serious risk of bias and serious imprecision.

Number of participants with new gadolinium-enhancing T1-weighted MRI lesions

Cladribine versus placebo may reduce the risk of developing new GD+ T1 MRI lesions, but the evidence is very uncertain (RR 0.68, 95% CI 0.52 to 0.88; 1 study, 806 participants; very low-certainty evidence; Analysis 10.6). This means that for every 1000 people receiving cladribine, 215 will develop new GD+ T1 MRI lesions, compared with 316 of 1000 people receiving placebo. We downgraded the evidence certainty for very serious risk of bias and serious imprecision.

Outcomes of limited importance

Number of participants with evidence of brain atrophy

No OLEs reported evidence of brain atrophy.

Cladribine compared to other disease-modifying drugs

Results from non-randomised studies of interventions

See [Summary of findings 5](#).

Both NRSIs reported data on relapses only, comparing cladribine with different DMDs (Kalincik 2018; Spelman 2023).

Important outcomes

Number of participants with at least one new relapse

Kalincik 2018 reported the number of participants experiencing at least one new relapse by one year of follow-up. Cladribine may have little to no effect on the risk of experiencing at least one new relapse by the end of follow-up compared with IFN β (RR 0.65, 95% CI 0.26 to 1.62; 1 study, 189 participants; very low-certainty evidence; Analysis 7.1), fingolimod (RR 1.05, 95% CI 0.52 to 2.10; 1 study, 290 participants; very low-certainty evidence; Analysis 5.1), and natalizumab (RR 1.25, 95% CI 0.58 to 2.71; 1 study, 200 participants; very low-certainty evidence; Analysis 6.1), but the evidence is very uncertain. We downgraded the evidence certainty in all three analyses for serious risk of bias and very serious imprecision.

Spelman 2023 reported the number of participants experiencing at least one new relapse by three years of follow-up. Cladribine may reduce the risk of experiencing at least one new relapse by the end of follow-up compared with dimethyl fumarate (RR 0.57, 95% CI 0.38 to 0.88; 1 study, 900 participants; low-certainty evidence; Analysis 8.1). We downgraded the evidence certainty for serious risk of bias and serious imprecision. Cladribine likely reduces the risk of experiencing at least one new relapse by the end of follow-up compared with fingolimod (RR 0.51, 95% CI 0.36 to 0.74; 1 study, 1040 participants; moderate-certainty evidence; Analysis 5.2), and likely results in a large reduction in the risk of experiencing at least one new relapse by the end of follow-up compared with teriflunomide (RR 0.30, 95% CI 0.20 to 0.47; 1 study, 916 participants; moderate-certainty evidence; Analysis 9.1). We downgraded the evidence certainty in both analyses for serious risk of bias.

Equity assessment

No included study reported sufficient data on equity-related aspects or disadvantaged subgroups. Consequently, we were unable to perform specific equity analyses or stratify results.

DISCUSSION

Summary of main results

In the critical safety outcome group, results from eight RCTs and two OLEs showed no evidence of a difference between cladribine and placebo in the risk of SAEs and treatment withdrawal by the end of study follow-up, but the evidence is very uncertain due to risk of bias and imprecision. Concern over malignancy risk – a key reason for licence refusal in 2010 [39] – was not confirmed in Giovannoni 2018 and Leist 2014 _OLMP, which found no increased incidence.

Based on three RCTs (low-certainty evidence) and one OLE (very low-certainty evidence), cladribine showed little or no effect on the number of people remaining free from disability worsening compared to placebo. Meta-analysis of two OLEs (low-certainty

evidence) suggested cladribine may increase the number of people with NEDA compared to placebo.

No data were available on cognitive impairment or QoL.

Among important outcomes, cladribine likely reduces the number of participants experiencing one or more new relapses compared to placebo (two RCTs, moderate-certainty evidence). Data from three OLEs (very low-certainty evidence) indicate that cladribine may reduce this outcome compared to placebo, but the evidence is very uncertain.

We are uncertain whether cladribine compared to placebo increases or reduces the risk of adverse events, as the evidence from eight RCTs and two OLEs was very uncertain owing to risk of bias, indirectness, and inconsistency.

Cladribine may reduce the number of participants with new Gd+ T1 MRI lesions at the end of follow-up compared to placebo (four RCTs and one OLE), but the evidence is very uncertain due to very serious risk of bias and inconsistency or imprecision.

Results from NRSIs were inconsistent due to the different sample sizes and follow-up periods considered. There is very uncertain evidence from Kalincik 2018 (one year of follow-up) about the effects of cladribine on the number of participants with at least one new relapse versus IFN β , fingolimod, or natalizumab. Low- to moderate-certainty evidence from Spelman 2023 (three years of follow-up) suggests cladribine likely reduces the risk of experiencing new relapses versus fingolimod, may reduce the risk versus dimethyl fumarate, and likely results in a large reduction in the risk versus teriflunomide.

Limitations of the evidence included in the review

Based on the study populations, we were unable to draw robust conclusions for planned subgroups. Most participants had RRMS (66.8%), followed by CIS (24.6%) and progressive MS (8.4%). Only early small trials assessed non-oral routes. No information was available for participants under 18 years or over 65 years, and prior DMD exposure was poorly reported.

Despite aiming for a comprehensive synthesis, several limitations emerged.

- The effect of cladribine beyond two years remains uncertain if based on RCT data, because the longest follow-up was 96 weeks. OLEs added a longer follow-up, but involved fewer, highly selected participants. Information on long-term effects could come from PREMIERE, a prospective, observational, long-term safety registry of people who had participated in at least one of the four trials, namely CLARITY, CLARITY-EXT, ONWARD, or ORACLE. We could not include key safety data because the PREMIERE study reported only aggregated results from pooled studies. Our request for disaggregated data from the Merck Group's Clinical Trial Transparency Team received no response [78].
- Reporting of key outcomes was incomplete. There were no data on some critical outcomes for patients and their families, such as cognitive impairment and QoL. Additionally, only a few trials used dichotomous analyses to assess disability worsening. We argue that dichotomous analyses provide more meaningful insights for both patients and clinicians compared with mean scores from neurological scales.

- The evidence relied heavily on one RCT (Giovannoni 2010) and its OLEs. Leist 2014 only considered CIS and was prematurely interrupted. A further study addressed safety and was amended due to side effects in the cladribine high-dose arm (Montalban 2018_original protocol). Moreover, we were unable to find the full report for Romine 1999, even after requesting it [98]. Overall certainty was low due to very high risk of bias.
- For most outcomes, we downgraded the evidence certainty for serious or very serious risk of bias and imprecision.
- Many studies were supported by Merck Serono; other older, small, independent studies contributed minimally to our results (Rice 2000; Romine 1999; Sipe 1994; Stelmasiak 2009).
- Overall, the evidence for cladribine remains insufficient for firm decision-making, as with other DMDs.

Limitations of the review processes

Some study authors published post-hoc analyses (e.g. MRI atrophy, QoL, NEDA) that were not prespecified in protocols, increasing bias and reducing validity; we therefore excluded these data.

We assessed risk of bias blindly using RoB 2. Applying this tool to older studies with limited methodological detail was challenging and may have led to overly strict judgements.

Agreements and disagreements with other studies or reviews

Compared with previously published reviews, our work adopted a broader search strategy, stricter eligibility criteria, no language restrictions, a different quality assessment, and a stronger focus on patient-centred outcomes. We identified three reviews on cladribine, each addressing different research questions [99, 100, 101]. Siddiqui and colleagues conducted a network meta-analysis comparing cladribine with other DMDs in active RRMS [99]; Oreja-Guevara and colleagues reviewed real-world evidence combined with expert opinion on long-term use [100]; and Nabizadeh and colleagues included RCTs and non-RCTs assessing safety and efficacy beyond six months and was the only review partially aligned with our methodology [101].

These reviews applied narrower search strategies, limited to major databases, and two of them provided little detail on study selection [100, 101]. The inclusion of mixed study designs, sometimes treating cohort studies as RCTs, further complicated quality appraisal. Study quality was assessed inconsistently: Siddiqui and colleagues used national agency criteria (NICE (National Institute for Health and Care Excellence), IQWiG (*Institut für Qualität und Wirtschaftlichkeit im Gesundheitswesen*), HAS (*Haute Autorité de Santé*)), rating all studies as low risk [99]; Oreja-Guevara and colleagues relied solely on expert opinion [100]; and Nabizadeh and colleagues applied the Newcastle-Ottawa scale for observational studies and the Cochrane tool for trials, judging half of studies at low risk [101], whereas we found most at high risk of bias.

Outcome measures also differed: Nabizadeh and colleagues focused on EDSS change, progression-free survival, and relapse-free survival [101]; Siddiqui and colleagues focused on relapse rate and NEDA [99]; and Oreja-Guevara and colleagues focused on disease progression and MRI activity [100].

Two reviews were industry sponsored, with company-employed co-authors, which may have influenced the interpretation of findings. [99, 100].

AUTHORS' CONCLUSIONS

Implications for practice

Cladribine is an oral short-course therapy (16 to 20 total treatment days over two years) with a low pill burden and limited monitoring needs.

In people with multiple sclerosis (MS), cladribine at a cumulative dose of 3.5 mg/kg versus placebo likely reduces the risk of experiencing at least one new relapse over 96 weeks of follow-up, and it may reduce the risk of developing new gadolinium-enhancing T1-weighted (Gd+ T1) magnetic resonance imaging (MRI) lesions over 12 to 22 months, but the evidence is very uncertain. Safety and tolerability appear similar to placebo, with no clear difference in serious adverse events or treatment discontinuations, though the evidence certainty is very low. The evidence on freedom from disability worsening was of low and very low certainty, showing only a non-significant trend in longer follow-up (Giovannoni 2018). Registry data comparing cladribine with other disease-modifying drugs (DMDs) were mixed: one study found no difference between the drugs in the risk of experiencing new relapses by one year, while another, larger study showed a reduction in favour of cladribine at three years.

Equity-related implications for practice

Cladribine offers key advantages for low-resource settings: it is oral, requires only two short annual courses, and needs no refrigeration or specialised training. Retreatment is unnecessary for at least two additional years, and monitoring needs are minimal. Pretreatment screening for latent infections (tuberculosis, hepatitis, HIV) is essential, and follow-up should include lymphopenia, infection risk (e.g. herpes zoster, pneumonia), and malignancy. The inclusion of cladribine on the World Health Organization (WHO) essential medicines list improves global access; however, affordability remains a key challenge for equitable implementation.

Implications for research

Several uncertainties remain, summarised as PICO questions.

There is no evidence for five specific populations: children and adolescents, pregnant women, female partners of men with MS [102], breastfeeding women [103], and older people. It is essential to understand the long-term consequences for developing immune systems, which are crucial for children and adolescents with a chronic disease. When treating older people, it is necessary to consider the potential for greater frequency of decreased organ function and other drug therapies. Future RCTs and large cohorts should include these groups, ensuring safety and representation. We also lack data on people previously treated with other DMDs, especially immunosuppressants. More head-to-head trials are needed to compare efficacy and long-term safety.

The outcome measures were too variable between trials, in type, measurement, and time. We found no data on outcomes such as quality of life and cognitive impairment. Future research should include these and explore disability improvement, not just worsening. In addition, the outcome measures were not focused on

the needs of people with a chronic disease. An important challenge highlighted by this meta-analysis is the need for researchers to use common outcomes and common ways to measure them (i.e. dichotomic versus continuous measures), making it easier for the results of studies to be compared and combined as appropriate. Moreover, future studies should improve the consistency and relevance of outcome measures, after discussion with patients and carers.

Finally, studies should clarify post-treatment strategies following a full cladribine cycle, establishing whether retreatment, switching, or escalation is most appropriate for people with inadequate response.

Equity-related implications for research

Most participants in cladribine trials were white. Evidence is lacking on whether safety or efficacy differ across racial or ethnic groups. Future studies should ensure greater diversity to improve the generalisability of findings.

SUPPLEMENTARY MATERIALS

Supplementary materials are available with the online version of this article: [10.1002/14651858.CD013524.pub2](https://doi.org/10.1002/14651858.CD013524.pub2).

Supplementary material 1 Search strategies

Supplementary material 2 Characteristics of included studies

Supplementary material 3 Characteristics of excluded studies

Supplementary material 4 Characteristics of ongoing studies

Supplementary material 5 Risk of bias

Supplementary material 6 Analyses

Supplementary material 7 Data package

Supplementary material 8 Hierarchy of outcomes according to their importance to assess the effect of cladribine on MS people

ADDITIONAL INFORMATION

Acknowledgements

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Editorial and peer-reviewer contributions

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The following people conducted the editorial process for this article.

- Sign-off Editor (final editorial decision): Graziella Filippini, Istituto Neurologico Carlo Besta, Milan.
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Contributions of authors

Maria Grazia Celani sadly passed away in March 2025. She conceived the idea for this review and remained actively involved in its development until the final days of her life, despite her declining health. She made substantial contributions to the conception, design, and drafting of the review. The remaining review authors completed the work, ensuring that her intellectual and scientific input was preserved and respectfully integrated into the final version.

MGC: conceptualisation, methodology, formal analysis, writing – original draft, writing – review & editing, validation.

History

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MO: methodology, formal analysis, data curation, investigation, writing – original draft, writing – review & editing.

MM: data curation, investigation, writing – original draft.

MVE: writing – original draft.

TAC: writing – original draft, writing – review & editing, supervision.

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Data, code and other materials

The following are made available for download for users of the Cochrane Library: full search strategies for each database; characteristics of all included, ongoing, or excluded studies; study data, including study information, study arms, and study results or test data; risk of bias assessments; and analysis data, including overall estimates, subgroup estimates, and individual data rows. Analyses and data management were conducted within Cochrane's authoring tool, Review Manager, using the inbuilt computation methods.

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ADDITIONAL TABLES

Table 1. Overview of included studies and synthesis

Study ID N° of countries/centres	Study design	Study arms ^a	Population	Key outcome domains: specific outcome measure	Time point of measures	Funding	Notes
Sipe 1994 Single centre	Phase III cross-over RCT	- Cladribine IV 2.8 mg/kg - Placebo	24 couples plus 1 person 67% women - EDSS 4.6 - PPMS and SPMS	- Clinical efficacy: disability worsening measured with EDSS - Safety: AEs - Instrumental: Gd+ T1 MRI lesions (local reading)	12 months	NIH grants, FDA grants, Stein Endowment Fund	Separate publication on MR outcome [27]
Romine 1999 Single centre	Parallel-group RCT	- Cladribine SC 2.1 mg/kg - Placebo	52 participants % women not reported - EDSS not reported - RRMS	Safety: AEs	18 months	NIH grant, J&J, Pharmaceutical Research, MS research	—
Rice 2000 2 countries, 6 centres	Parallel-group RCT	- Cladribine SC 2.1 mg/kg - Cladribine SC 0.7 mg/kg - Placebo	159 participants 67.7% women - EDSS 5.6 - PPMS (30.3%) and SPMS	- Safety: AEs - Instrumental: Gd + T1 MRI lesions (central reading)	12 months	Not declared	—
Janiec 2001 Single centre	Cross-over RCT	- Cladribine SC 2.1 mg/kg - Placebo	69 participants 55.1% women - EDSS not reported - RRMS and CPMS	Safety: AEs	24 months	Drug supply by Therapeutic Foundation	—
Stelmasiak 2009 Single centre	Cross-over RCT	- Cladribine SC 2.45 mg/kg - Placebo	84 participants 58.3% women - EDSS 3.9 - RRMS, naïve	Safety: AEs	12 months	Development of Diagnostic and Therapy Warsaw	—
Giovannoni 2010 32 countries, 155 centres	Phase IIIB parallel-group RCT	- Cladribine O 3.5 mg/kg - Cladribine O 5.25 mg/kg - Placebo	1326 participants 67.7% women - EDSS 2.9 - RRMS	- Clinical: relapse; disability worsening measured with EDSS - Safety: AEs	96 weeks (22 months)	MerkSerono	Separate publication on MR results [32]

Table 1. Overview of included studies and synthesis (Continued)

				Instrumental: Gd + T1 MRI lesions (central reading)			
Leist 2014 34 countries, 160 centres	Phase III parallel-group RCT	- Cladribine O 3.5 mg/kg - Cladribine O 5.25 mg/kg - Placebo	616 participants 65% women - EDSS 1.5 - CIS	- Safety: AEs - Instrumental: Gd + T1 MRI lesions (central reading)	< 96 weeks (22 months) for early trial termination	MerkSerono	—
Leist 2014 _OLMP 34 countries, 160 centres	Open-label study (maintenance period)	Participants started IFN- β 1a, 44 μ g SC as standard treatment	109 participants entered the maintenance period.	Safety: AEs	96 weeks (22 months)	MerkSerono	Subgroup who converted to MS during Leist 2014
Giovannoni 2018 30 countries, 133 centres	Partially randomised extension trial	Interventions in CLARITY study + CLARITY Extension: - Cladribine O, 3.5 mg/kg + placebo - Cladribine O, 5.25 mg/kg + placebo - Cladribine O, 3.5 mg/kg + 3.5 mg/kg - Cladribine O, 5.25 mg/kg + 3.5 mg/kg - Placebo + Cladribine O, 3.5 mg/kg	806 participants who completed Giovannoni 2010 65.9% women - EDSS 2.6	- Clinical: relapse; disability worsening measured with EDSS - Safety: AEs - Instrumental: Gd + T1 MRI lesions (central reading)	96 weeks (22 months)	EMD, MerkSerono	Separate publication on MR results [40]
Montalban 2018_original protocol 4 countries, 50 centres	Parallel-group RCT (original protocol)	Treatment added to IFN- β SC: - Cladribine O 3.5 mg/kg - Cladribine O 5.25 mg/kg - Placebo	42 participants 69.7% women - EDSS 2.9 - RRMS (90.5%) and SPMS	- Clinical: relapse - Safety: AEs	< 96 weeks (22 months)	MerkSerono	Original study terminated early
Montalban 2018_amended 4 countries, 50 centres	Parallel-group RCT (amended protocol)	Treatment added to IFN- β SC: - Cladribine O 3.5 mg/kg - Placebo	172 participants 69.7% women - EDSS 2.9 - RRMS (90.5%) and SPMS	- Clinical: relapse; disability worsening measured with EDSS - Safety: AEs	< 96 weeks (22 months)	MerkSerono	—

Table 1. Overview of included studies and synthesis (Continued)

Kalincik 2018 20 countries, 45 centres	4 cohorts of participants included in registers were matched by propensity score analysis	• Cladribine O 3.5 mg/kg • Fingolimod O • Natalizumab IV • IFN-β SC	• 22 cladribine vs 167 IFN β; 32 cladribine vs 258 fingolimod; 26 cladribine vs 174 natalizumab • 76.6% women • EDSS 3.8 • RRMS (84.8%) and SPMS (15.2%)	• Clinical: relapse; disability worsening measured with EDSS	12 months	Australian National Grant, MSBase Foundation (received multiple drug industry supports)	—
Spelman 2023 31 countries (MSBase registry)	Longitudinal, real-world observational registry study (secondary use of prospectively collected data)	• Cladribine O • Fingolimod O • Dimethyl fumarate O • Teriflunomide O	• Cladribine: 633; fingolimod: 1195; dimethyl fumarate: 912; teriflunomide: 735 • > 70% women in all groups • > 87% RRMS in all groups	• Clinical: annualised relapse rate (ARR) and time to first relapse (propensity score-matched analyses; ARR via Poisson models)	3 years	Supported by EMD Serono (an affiliate of Merck KGaA, Darmstadt, Germany)	—
Giovannoni 2023 29 countries, 98 centres	Multicentre, ambispective, low-interventional, phase IV exploratory study	Follow-up of participants from CLARITY/CLARITY Extension who had received ≥ 1 course of cladribine tablets (3.5 mg/kg cumulative dose over 2 years) or placebo	• 435 participants from CLARITY (with/without CLARITY Extension); 394 exposed to cladribine (160 with 3.5 mg/kg dose), 41 placebo only • Median age 52.5 years • 67.8% women • Median disease duration 22.4 years • Median time since last dose 10.9 years	• Clinical: relapse	Median 10.9 years since last parent study dose (range 9.3–14.9 years)	Sponsored by Merck	—
Giovannoni 2025 29 countries, 98 centres	Phase IV ambispective (retrospective + prospective) observational exploratory study (CLASSIC-MS, long-term follow-up of ORACLE-MS)	Participants previously treated with ≥ 1 course of oral cladribine (3.5 mg/kg or 5.25 mg/kg) or placebo in ORACLE-MS; no new dosing during CLASSIC-MS	• 227 participants (156 exposed, 71 unexposed) • Median age 41.5 years • 65.6% women • Initial diagnosis FCDE/CIS • Median follow-up 9.5 years after last dose	• Clinical: relapse	Median 9.5 years (range 8.2–11.2 years) after last dose in ORACLE-MS	Merck KGaA	—

^a All doses of cladribine are cumulative.

AE: adverse event; **CIS:** clinically isolated syndrome; **EDSS:** Expanded Disability Status Scale; **EMD Serono:** healthcare business of Merck KGaA; **FCDE:** first clinical demyelinating event; **FDA:** US Food and Drug Administration; **Gd+ T1 MRI lesions:** gadolinium-enhancing T1-weighted magnetic resonance imaging lesions; **IFN-β:** Interferon beta; **IV:** intravenous; **MS:** multiple sclerosis; **NIH:** National Institute for Health; **O:** oral; **PPMS:** primary-progressive multiple sclerosis; **RRMS:** relapsing-remitting multiple sclerosis; **RCT:** randomised controlled trial; **SC:** subcutaneous; **SPMS:** secondary progressive multiple sclerosis.

Table 2. ROBINS-I: review authors' judgements about each risk of bias domain (Kalincik 2018)

Outcome	Interven- tion effect	Domains							Overall risk of bias
		Bias due to con- founding	Bias in selec- tion of partici- pants into the study	Bias in clas- sification of interven- tions	Bias due to deviations from intend- ed interven- tions	Bias due to missing data	Bias in measure- ment of outcomes	Bias in se- lection of the re- ported re- sult	
Number of participants who ex- perienced ≥ 1 new relapse over 12 months after randomisation and at the end of follow-up	Assign- ment	Moderate ^a	Serious ^b	Moderate ^c	Low ^d	Moderate ^e	Serious ^f	Low ^g	Serious

^a Bias due to confounding by indication expected. Important confounding domains were appropriately measured and controlled for. Author stated "Individual propensity scores were calculated using a multivariable logistic improvement model of treatment allocation that used as independent variables the demographic and clinical variables available at the time of treatment assignment: sex, age, time from first MS symptom, EDSS, MS course, number of relapses in the prior 3 months, number of prior MS therapies and the most effective prior MS therapy".

^b Start of follow-up and start of intervention do not coincide for all participants, important amount of follow-up is missing from the analysis, and the rate ratio is not constant over time.

^c Retrospective clinical data were collected from the registries. The intervention groups were clearly defined.

^d Any deviations from the intended interventions (not reported) were part of usual clinical practice.

^e Proportions of missing participants differ slightly across intervention groups (see Kalincik 2018, Supplementary Table S3), while reasons are not clearly described. The analysis is unlikely to have removed the risk of bias arising from the missing data. Authors stated: "The common on-treatment follow-up was determined in each matched pair as the shorter of the 2 patient follow-up periods (pairwise censoring) to mitigate attrition bias, informative censoring and the effect of differential treatment persistence".

^f Outcome assessors were aware of the intervention received by study participants, which could have affected their judgement.

^g Although reported results corresponded to outcomes listed in the methods section of the article, the choice to define disability in 2 outcomes (disability accumulation and disability improvement through EDSS) is questionable (we decided to include only the disability accumulation). There is a low risk of selective reporting among primary analysis (but there are numerous sensitivity analyses).

EDSS: Expanded Disability Status Scale; **MS:** multiple sclerosis.

Table 3. ROBINS-I: review authors' judgements about each risk of bias domain (Spelman 2023)

Outcome	Interven- tion effect	Domains							Overall risk of bias

Table 3. ROBINS-I: review authors' judgements about each risk of bias domain (Spelman 2023) (Continued)

		Bias due to confounding	Bias in selection of participants into the study	Bias in classification of interventions	Bias due to deviations from intended interventions	Bias due to missing data	Bias in measurement of outcomes	Bias in selection of the reported result	
Number of participants who experienced ≥ 1 new relapse over 12 months after randomisation and at the end of follow-up	Assignment	Serious ^a	Serious ^b	Low ^c	Low ^d	Serious ^e	Serious ^f	No information ^g	Serious

^a Although 1:1 propensity-score matching achieved balance (standardised mean difference < 0.15) on multiple key baseline determinants, the non-randomised design and absence of important prognostic domains (most notably baseline MRI activity acknowledged by the authors) make residual and unmeasured confounding likely. Given confounding by indication and centre/care patterns cannot be excluded in this real-world registry, bias due to confounding remains serious for the binary relapse outcome at a fixed horizon.

^b Inclusion in the analytic cohorts required postindex characteristics (most notably > 6 months of follow-up and availability of a minimum dataset), so selection depended on variables observed after treatment start. These postintervention variables (persistence/availability of follow-up) are plausibly associated with the intervention initiated (the study reports different persistence profiles across cohorts) and with relapse risk (switch/discontinuation often triggered by disease activity). Although follow-up was anchored at the index date, no methods were used to correct selection on postindex characteristics (e.g. inverse-probability selection weighting). Therefore, selection into the study/analysis is likely related to both intervention and outcome, introducing serious risk of bias.

^c Interventions were clearly and operationally defined as initiation of cladribine, fingolimod, dimethyl fumarate, or teriflunomide, with the index date = treatment start and follow-up beginning at index, so classification relied on information captured at the time of intervention (before the outcome). Switches/discontinuations did not trigger reclassification of exposure but pairwise censoring, limiting misclassification over time; the same registry procedures were applied across cohorts, making differential misclassification unlikely.

^d This was an observational registry study conducted in routine clinical practice (MSBase). Postinitiation changes such as nonadherence, switching, and discontinuation reflect usual care rather than trial-context deviations; no protocol-driven behaviours that would not occur outside routine practice were reported. Therefore, deviations beyond usual practice are not evidenced, and questions about imbalance and impact are not applicable.

^e Outcome data at a fixed 3-year horizon were not available for all or nearly all participants; inclusion required > 6 months of postindex follow-up and a minimum dataset for propensity-score derivation, leading to exclusions for missing confounders and variable follow-up across cohorts. Proportions and likely reasons for missing data differed by index treatment (reflecting distinct persistence/switching patterns), and no corrective methods or sensitivity analyses were reported to assess robustness. Together, these features indicate a serious risk of bias due to missing data.

^f Relapses were identified from routine clinical documentation in an unblinded, real-world setting; determining whether an event qualified as a relapse required clinician judgement. Although the registry applied the same operational definition across cohorts, treating clinicians knew which DMT had been initiated and monitoring/persistence patterns differed between groups, making differential detection and misclassification of relapse events plausible. These features could materially bias the estimated proportion with ≥ 1 relapse by end of follow-up.

^g For the binary endpoint 'number of participants who experienced at least 1 new relapse over 12 months after randomisation and at the end of follow-up', there is no time-stamped protocol/SAP specifying this measure, its analytic approach, or subgrouping; the estimate was provided via author correspondence and is not reported in the paper. Thus, there is insufficient information to judge whether the reported effect was selected from multiple outcome measurements, analyses, or subgroups.

DMT: disease-modifying therapy; **MRI:** magnetic resonance imaging; **MSBase:** MSBase registry; **SAP:** statistical analysis plan.